Infections & Inflammation

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 26	Infections of the Urinary Tract
Ch 27	Prostatitis, Orchitis & Pain Conditions
Ch 28	Interstitial Cystitis / Bladder Pain Syndrome
Ch 29	Sexually Transmitted Diseases
Ch 30	Cutaneous Diseases of the External Genitalia
Ch 31	Tuberculosis & Parasitic Infections
★	Putting it together: infection decisions

26 Infections of the Urinary Tract

1 · Definitions and classification

Why this matters: the label (complicated or not, reinfection or persistence) decides treatment length and whether to investigate.

- **Urinary tract infection (UTI)** is the inflammatory response of the urothelium to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**, and bacteriuria may even protect.
- **Bacteriuria** (colonisation, infection or contamination) and **pyuria** (white cells from infection or from inflammation such as a stone or foreign body) are not the same as UTI. Bacteriuria without pyuria is colonisation. **Sterile pyuria**: look for tuberculosis (TB) *see Ch 31*, stone, urothelial injury and interstitial nephritis.
- **Cystitis** causes dysuria, frequency and urgency, with or without suprapubic pain. Its mimics are urethritis, vaginitis, interstitial cystitis/painful bladder syndrome (IC/PBS) *see Ch 28*, bladder carcinoma and calculi.
- **Acute pyelonephritis** is chills, fever and flank pain with bacteriuria and pyuria. It is a clinical diagnosis, not made on perinephric stranding alone. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis** shows a focal coarse cortical scar over a **distorted calyx**, identical to vesicoureteral reflux (VUR) or calyceal-stone scarring. Uniform thinning suggests obstruction with infection.

	Uncomplicated UTI	Complicated UTI
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Often multidrug-resistant (MDR), so culture is mandatory

- **Complicating factors** are an anatomical or functional abnormality (benign prostatic hyperplasia [BPH], diverticula, ureteric obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO], concentrating defects), **male sex**, pregnancy, old age, diabetes mellitus (DM), immunosuppression, childhood UTI, recent antibiotics, catheter, instrumentation, hospital acquisition and symptoms >7 days. MCQ Male sex alone makes a UTI complicated.
- US Food and Drug Administration (FDA) 2018: complicated UTI is pyuria, a pathogen in urine or blood, and local or systemic features (fever >38 °C, flank pain, costovertebral angle [CVA] tenderness) in the presence of an abnormality or catheter.

Term	Definition	Discriminator
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New inoculation (~**70%** of recurrences, often different organism)	Cleared, then new event
Bacterial persistence (relapse)	Same organism from a focus (stone, prostate)	Focus remains; recurs quickly

- **Prophylaxis** prevents reinfection of a sterilised tract; **suppression** prevents growth from an ineradicable focus.
- **Catheter-associated UTI (CAUTI)** is the commonest cause of secondary bacteraemia and accounts for ~**80%** of nosocomial UTIs.
- Recurrence risk rises with the number of previous UTIs and falls with time since the last. Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · Pathogenesis

Why this matters: how bacteria enter and hide explains why cystitis recurs and why obstruction is dangerous.

Infection depends on virulence and inoculum weighed against host defence.

Route	Frequency	Features
Ascending	Commonest	Bowel and skin flora colonise the introitus and climb to bladder and ureter. Helped by faecal soiling, spermicide, catheters and antiperistaltic factors (endotoxin, pregnancy, obstruction). Reflux not required.
Haematogenous	Uncommon	*S. aureus*, *Candida*; enhanced by obstruction
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- ***E. coli*** causes ~**85%** of community and ~50% of hospital UTIs. Its share falls from ~¾ in uncomplicated cystitis to <½ in complicated UTI and urosepsis.
- ***S. saprophyticus*** causes ~10% of cystitis in young sexually active women, and only in uncomplicated cystitis.
- Nosocomial and complicated UTIs: *Klebsiella*, *Enterobacter*, *Serratia*, *Pseudomonas* and *E. faecalis*. Culture-negative chronic symptoms: *Ureaplasma* or *M. hominis*.
- Uropathogenic *E. coli* (UPEC) are mostly phylogroup B2. **ST131 (O25b:H4)** is an MDR clone that carries extended-spectrum β-lactamase (ESBL); fluoroquinolone resistance is its hallmark.
- MCQ *Enterococcus* is a Gram-positive coccus, not an Enterobacteriaceae.
- **Anaerobes** are rare in cystitis, but are found in most scrotal, prostatic and perinephric abscesses in men and in enteric fistulae. Suspect them when microscopy shows organisms but aerobic culture is sterile.
- Urinary TB presents with haematuria or sterile pyuria; polymerase chain reaction (**PCR**) has replaced acid-fast staining.

Virulence — fimbriae

- UPEC virulence genes sit on pathogenicity-associated islands (PAIs): adhesins, toxins (HlyA, Sat), capsule, flagella and iron acquisition. Adherence is the essential first step.

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	*fim*; **FimH** tip adhesin	*pap* operon on a PAI
Receptor	Mannosylated **uroplakins Ia/Ib**	Glycolipid (P blood-group) receptors in kidney

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Tropism	Bladder, causing cystitis	Kidney, causing **pyelonephritis** (~**90%** of pyelonephritis strains)

- P pili are found most often in pyelonephritis strains, then cystitis, then asymptomatic bacteriuria (ASB), then faecal isolates. In **phase variation** bacteria switch between piliated and non-piliated states. Type 1 pili promote phagocytosis once bacteria float free, so P pili take over in the kidney.

Intracellular cycle (why cystitis recurs)

ALGORITHM

1. FimH binds uroplakin and the bacterium invades the umbrella cell (15 min–1 h).
2. It multiplies into an **intracellular bacterial community (IBC)**, which behaves like a biofilm (4–16 h).
3. Bacteria filament, escape and reinvade neighbouring cells (16–24 h).
4. Infected cells exfoliate, exposing deeper layers.
5. **Quiescent intracellular reservoirs (QIRs)** form (from ~2 weeks).

- QIRs are inert and silent between attacks. Urothelial disruption reactivates them, giving same-strain recurrence without a urological focus. Biofilm shields bacteria from antibiotics and neutrophils.

Host defences and susceptibility

- **Introital receptivity**: *E. coli* adheres more to vaginal cells of women with recurrent cystitis. This is genetic and linked to Lewis blood-group phenotypes.
- Lactobacilli are the barrier; antimicrobials, spermicides and oestrogen loss (pH rises) disturb them. **Uromodulin (Tamm–Horsfall protein)**, the most abundant urinary protein, competes with uroplakins for type 1 pili.
- Bladder emptying is the key defence; **obstruction** makes infection life-threatening.
- **VUR**: in children, high-grade VUR with UTI leads to scarring and renal failure. In adults VUR alone is harmless without high pressure or stasis plus UTI.
- **DM** brings more pyelonephritis, abscess, emphysematous infection, fungal infection and papillary necrosis. In **human immunodeficiency virus (HIV)** infection UTI is ~5× commoner; with CD4 <200, expect atypical organisms. Trimethoprim–sulfamethoxazole (TMP-SMX) given as Pneumocystis prophylaxis also reduces UTI. After **renal transplant** >⅓ get UTI, and graft pyelonephritis predicts functional decline. **SCI** carries the highest morbidity of any bacteriuric group.

Renal papillary necrosis

- The main causes are **DM, obstruction with pyelonephritis, analgesic abuse and sickle cell disease**; also rejection, cirrhosis, dehydration, renal vein thrombosis and candidiasis.
- A sloughed papilla obstructs the ureter; retained papillae calcify into a nidus. Treat with antibiotics and early relief of obstruction.

3 · Evaluation and diagnosis

Why this matters: most UTIs need only history and a urine test; the skill is spotting mimics and knowing when culture or imaging helps.

Syndrome	Features
Cystitis	Dysuria, frequency, urgency, suprapubic pain, haematuria; afebrile
Pyelonephritis	Fever, chills, flank pain, CVA tenderness, with or without vomiting; lower urinary tract symptoms (LUTS) may be absent
Fistula	Pneumaturia, faecaluria
Renal/perirenal abscess	Indolent fever, flank mass
Catheter	Usually ASB; bacteraemia can develop fast

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues / causes	*E. coli*	Discharge, odour, new partner; herpes simplex virus (HSV), gonorrhoea, *Chlamydia*, *Trichomonas*, yeast, bacterial vaginosis	Discharge; initial-stream pyuria in men; *N. gonorrhoeae*, *Chlamydia*, HSV *see Ch 29*

- Dysuria with frequency and no vaginal discharge gives a ~**90%** probability of UTI; haematuria with a positive nitrite gives ≥90%. Dysuria is the best single symptom.
- **Rule**: painless gross haematuria, or microhaematuria with a negative culture or persisting after the infection clears, needs full upper and lower tract evaluation.

Urine collection

- Midstream urine is the default, though often contaminated in women. An in-and-out catheter sample is more accurate and halves unnecessary prescribing; a suprapubic aspirate is the most accurate.
- Indwelling catheter: replace it and sample the new one. Stoma: catheterise it, never sample the bag.

Urinalysis and culture

- **Leucocyte esterase** indicates pyuria. **Nitrite** is made by Enterobacteriaceae; it is negative with Gram-positives, *Pseudomonas*, *Acinetobacter*, a low-nitrate diet or a short bladder dwell. High specificity and low sensitivity make nitrite the best rule-in test. Dipsticks are better at **ruling out**, and a negative automated urinalysis excludes UTI.
- On microscopy, pyuria is ≥**5–10 WBC/high-power field (HPF)**, and squamous cells mean contamination. No pyuria: question the diagnosis. Positive culture without pyuria: repeat on a catheter sample.

Threshold	Use
≥10^5 colony-forming units (CFU)/mL	Kass (originally pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheter specimens

- Counts are low in cystitis because frequent voiding washes bacteria out before 10^5.
- An "I" (intermediate) susceptibility may still work, because urine levels are often several hundred times serum levels. All common oral agents reach inhibitory urine levels except macrolides.

- **Microbiome** studies (sequencing, expanded quantitative urine culture [EQUC]) show that "no growth" does not mean sterile and an organism does not mean infection. Sequencing gives no susceptibilities, so decide on symptoms.
- Localise a focus by **ureteric catheterisation (Stamey)** for one kidney, an ectopic ureter or a ureteric stump. **Stone culture** predicts post-procedure sepsis and systemic inflammatory response syndrome (SIRS) better than bladder urine.

Imaging

Imaging asks only: is there obstruction or a collection to drain, or a focus of persistence? Most UTIs need none.

- **Indications in pyelonephritis**: possible obstruction (stone, stricture, tumour), risk of papillary necrosis, prior reconstruction, **no response after 72 h**, DM, polycystic kidneys on dialysis, NGB, and unusual organisms (TB, fungi, urea-splitters). Also image most men, compromised hosts and rapid recurrences.

Question	Best test
Stone	Non-contrast computed tomography (CT)
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Hydronephrosis, pyonephrosis, post-void residual (PVR)	Ultrasound (US)
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG): febrile UTI in women, childhood VUR, recurrent adult pyelonephritis
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6)
Infection stones; chronic bacterial prostatitis; unilateral infected atrophic kidney; duplication/ectopic ureter; foreign body; urethral diverticulum/periurethral glands; unilateral medullary sponge kidney; non-refluxing infected ureteric stump; infected urachal cyst; infected communicating calyceal cyst; papillary necrosis; perivesical abscess with fistula. Antibiotics cannot sterilise these, so **cure comes only by removal or drainage.**

4 · Antimicrobial principles

Why this matters: the drug depends on where the infection sits, and every course breeds resistance.

- In cystitis the **urinary** drug level matters. In parenchymal infection or bacteraemia, **serum and tissue** levels matter. Renal insufficiency and obstruction lower urine levels, so adjust renally cleared drugs.
- In cystitis the goal is symptom relief; non-steroidal anti-inflammatory drugs (NSAIDs) alone give longer symptoms and more pyelonephritis.
- **Collateral damage** means resistance, MDR colonisation, *C. difficile* and candidiasis, caused mainly by broad cephalosporins and quinolones.
- P1 Antimicrobial resistance; ESBL producing enterobacterial UTI — *intrinsic / chromosomal / plasmid resistance; ESBL, AmpC, KPC and their treatment options*

Not in the paper answer:

- Resistance scales with use, and reducing use can restore susceptibility.
- Chromosomal resistance is prevented by full dosing, good compliance and a drug whose urine level far exceeds the minimum inhibitory concentration (MIC).
- *Klebsiella pneumoniae* carbapenemase (KPC) is the commonest US carbapenemase and is spreading to other Gram-negatives. AmpC, once chromosomal, is now also plasmid-borne.
- Stewardship programmes shorten courses.
- Plasmid resistance spreads among **bowel** flora; one drug selects resistance to many, so post-antibiotic UTIs are often MDR. Prior quinolone use is the strongest determinant of quinolone resistance.
- **Stewardship** means narrow spectrum, the shortest course, culture before treating recurrent or complicated UTI, and use of local antibiograms.

Agents

Class	Action	Resistance
β-lactams	Cell wall	β-lactamase, altered penicillin-binding protein (PBP), porins
Fosfomycin	Cell wall, at a different step, so there is little cross-resistance	Transporter loss
Vancomycin	Cell wall	Altered peptidoglycan
Aminoglycosides	Ribosome, bactericidal	Modifying enzymes
Quinolones	DNA gyrase	Gyrase mutation, efflux, porins
TMP-SMX	Folate antagonism	Enterococci scavenge folate (fail even if "sensitive")
Nitrofurantoin	Multiple enzymes	Very low

- **Anti-pseudomonal** agents are antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides and fluoroquinolones.
- **Enterococcal** cover comes from aminopenicillins, piperacillin, nitrofurantoin and fosfomycin, and **never from cephalosporins**.
- Aztreonam and pivmecillinam have no Gram-positive cover; vancomycin has no Gram-negative cover. Ertapenem misses *Pseudomonas* and *Enterococcus*; carbapenems miss *E. faecium* and methicillin-resistant *S. aureus* (MRSA).

Agent	Key points
Nitrofurantoin	Acts in urine only, so it is useless for pyelonephritis, prostatitis and bacteraemia. Spares bowel flora. Toxicity: pulmonary fibrosis, neuropathy, haemolysis in glucose-6-phosphate dehydrogenase (G6PD) deficiency. Beers 2015: avoid if creatinine clearance (CrCl) < **30 mL/min** (older advice 50–60); no long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses *Enterococcus*, *Pseudomonas*. Commonest antibiotic cause of Stevens–Johnson syndrome/toxic epidermal necrolysis (SJS/TEN); potentiates warfarin.
Fosfomycin	3 g single dose; active against ESBL producers and vancomycin-resistant enterococci (VRE), so it is an oral option for MDR cystitis.
Fluoroquinolones	Broad, including *Pseudomonas*. Adjust at CrCl <30. FDA warnings: tendon rupture (age >60, steroids, renal failure), neuropathy, dysglycaemia, psychiatric effects, aortic dissection; **2016: reserve for when there is no alternative in uncomplicated UTI**. Avoid in children and pregnancy. Separate from antacids and iron by ≥2 h; interacts with theophylline and warfarin.

Agent	Key points
Cephalosporins	Poor against enterococci; 3rd generation is the most reliable β-lactam for Gram-negatives. Use in cystitis only if first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once daily 5–7 mg/kg (concentration-dependent killing, post-antibiotic effect, less nephrotoxicity). A nomogram level 6–14 h after the dose sets the interval at 24, 36 or 48 h. Monitor oto- and nephrotoxicity.
Aztreonam	Aminoglycoside-like spectrum without nephrotoxicity; useful in penicillin allergy and renal impairment.
Pivmecillinam	~2% *E. coli* resistance; Nordic first-line; not in USA.

KEY POINT

Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis; broad tissue-penetrating agents (quinolones, cephalosporins, carbapenems) are kept for invasive infection. Resistance is made in the bowel, so breadth, duration and repetition matter more than MIC.

C. difficile infection (CDI)

- The usual culprits are cephalosporins, clindamycin, carbapenems, quinolones and TMP-SMX. Diagnosis needs ≥3 unformed stools in 24 h and toxigenic *C. difficile* or its toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin, with or without **bezlotoxumab** (anti-toxin B antibody)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily and IV metronidazole

5 · Asymptomatic bacteriuria

Why this matters: treating ASB usually harms, so know the few groups who need it.

Group (Infectious Diseases Society of America [IDSA])	Specimen	Threshold
Women	**Two** consecutive voided, same strain	≥10^5 CFU/mL
Men	One voided	≥10^5 CFU/mL
Either sex	One catheterised	≥ **10^2** CFU/mL

- Prevalence is 1–5% in premenopausal women, 25–50% in long-term care and **100%** with a long-term catheter.
- Untreated ASB causes no hypertension, renal failure, cancer or recurrent UTI. Treating it brings more symptomatic UTI, MDR and side effects; ASB strains may protect (bacterial interference).

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; elderly (community/institution); SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (incl. arthroplasty, cardiac)**	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection [TUR], ureteroscopy, percutaneous nephrolithotomy [PCNL]). No consensus: neutropenia.

KEY POINT

ASB is treated only where bacteria are about to reach blood or tissue: pregnancy (hydroureter leading to pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · Cystitis

Why this matters: a short narrow course cures most women; men and complicated cases need longer treatment and a cause.

Acute uncomplicated cystitis in women

- Risk factors, grouped by defence defeated: **washout** (obstruction, NGB, dehydration); **flora and binding** (intercourse, spermicide, oestrogen loss, antibiotics); and **ascent** (catheter, incontinence, residual urine).
- Diagnose on acute dysuria with a change in voiding. Fever, flank pain or no oral route: treat as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance >**20%** or used in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Ecology and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

A **3-day** course beats a single dose and equals 7 days (7 if complicating factors). No post-treatment culture if asymptomatic.

Cystitis in men

- Male cystitis is complicated; about half have an abnormality (incomplete emptying), so evaluate. Consider prostatitis *see Ch 27*.
- In afebrile men, **7 days is as good as 14 days** (Drekonja). The European Association of Urology (EAU) suggests TMP-SMX 160/800 twice daily × 7 d, or a quinolone guided by local susceptibility.

Complicated UTI

Category (Canadian Urological Association [CUA])	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction (PUJ) obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

Always culture and correct the abnormality. Treat **10–14 days** by culture, IV to oral once afebrile; re-culture if no response.

Emphysematous cystitis

- Bladder-wall gas, typically in an elderly diabetic woman. *E. coli* causes ~60%, then *Klebsiella*. Abdominal pain (~80%) and gross haematuria are usual, but fever occurs in only 30–50%. A plain film shows a curvilinear lucency; **CT confirms** it and excludes a fistula.
- ~**90%** settle medically with broad Gram-negative cover, catheter drainage and glucose control; cystectomy or debridement is rarely needed.

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection, new resistant species	New species; suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn	Same species, still **sensitive**	Renal function and imaging
8. Non-compliance, Munchausen	—	—

7 · Recurrent UTI

Why this matters: most recurrences are reinfections, so the work is prevention, not investigation.

- Recurrent UTI means **2 in 6 months or ≥3 in 12 months**, each culture-documented; ~¼ of young women recur within 6 months.
- Reinfection is a different organism, or the same one >2 weeks after treatment; it is commoner and arises from faecal flora or QIRs. In men, reinfection also suggests an anatomical abnormality.
- Risk factors: intercourse, new partner, spermicide, affected first-degree female relative, premenopausal UTI, menopause, recent antibiotics, incontinence, cystocele, raised PVR.
- Evaluate with history (pneumaturia, faecaluria, diverticulitis, radiotherapy), pelvic examination, **culture of each episode**, and PVR with or without flow rate. **Imaging and cystoscopy are not routine** (American Urological Association [AUA], EAU, CUA).

Indications for imaging with or without cystoscopy in recurrent UTI (CUA)
Prior urinary surgery/trauma; stones; gross haematuria or persisting microhaematuria; obstructive symptoms, low flow, high PVR; urea-splitters; prior abdominopelvic malignancy; persistence despite sensitivity-based therapy; DM/immunocompromise; pneumaturia, faecaluria, anaerobes, diverticulitis; repeated pyelonephritis. MCQ Recurrence frequency alone is **not** an indication.

- **Struvite** stones form when urease (from *Proteus mirabilis*) produces magnesium ammonium phosphate, and the bacteria live inside the stone. The AUA advises complete removal of infection stones in fit patients; unfit patients get acetohydroxamic acid and suppression.

Prevention (women, AUA 2019)

ALGORITHM

1. Confirm recurrent UTI with symptoms and culture.
2. If the same organism returns at short intervals, then evaluate for persistence (upper tract imaging and cystoscopy) and remove the nidus.
3. If it is reinfection, then start behavioural measures: ≥1.5 L water a day, frequent and postcoital voiding, stop spermicide.
4. Add cranberry ≥36 mg proanthocyanidin (PAC).
5. If UTIs follow intercourse, then give a **postcoital antibiotic**. If postmenopausal, then give **vaginal oestrogen**. If neither, then consider **methenamine hippurate** (normal renal function).
6. Use continuous antibiotic prophylaxis last.

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	Randomised controlled trial (RCT) in premenopausal women: 1.7 vs 3.2 episodes/yr
Vaginal oestrogen	Restores lactobacilli, lowers pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm); benefit after ≥12 weeks. **Oral oestrogen ineffective.** No excess breast cancer/cardiovascular risk; breast cancer survivors with oncologist agreement.
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable; safe with warfarin
Methenamine hippurate 1 g twice daily	**Formaldehyde in acid urine (pH <6)**	Non-inferior to daily prophylaxis. Fails with urease producers, NGB or anatomical abnormality. Contraindicated if CrCl <50 or liver disease; antacids, carbonic anhydrase inhibitors and citrus reduce effect.
D-mannose	Inhibits **FimH**	Cochrane 2022 inconclusive
Immunostimulants (OM-89 oral, Urovac vaginal, Uromune MV140 sublingual)	Bacterial lysates	Some benefit; small biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Works only while taken; MDR	AUA "may prescribe"; EAU: last resort

Antibiotic strategy	Evidence	Position
Postcoital single dose	Effective; preferred for coital UTI	Acceptable
Self-start	Old data; overuse	Prefer analgesic, sample and observe, then culture-directed treatment

KEY POINT

Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigate only for features of persistence or complication; prevention targets flora, pH, adhesion and washout rather than repeatedly sterilising urine.

8 · Kidney infections

Why this matters: some renal infections settle on tablets, others kill unless drained; imaging tells them apart.

Acute pyelonephritis

- Onset is abrupt: chills, fever ≥38 °C, flank or CVA pain. LUTS are often absent; it may mimic bowel disease. Bloods show leucocytosis and raised C-reactive protein (CRP); **white cell casts** prove a renal origin. Take blood cultures if the patient is toxic, admitted or pregnant.
- The usual cause is P-fimbriated *E. coli*. Suspect resistance after recurrent UTI, hospital stay, catheter or instrumentation.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%**; levofloxacin 750 mg daily; TMP-SMX double-strength twice daily only if known sensitive. With or without one IV dose of ceftriaxone 1 g or an aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin and gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci: ampicillin–sulbactam with or without an aminoglycoside.	10–14 total
Pregnancy	Ampicillin and gentamicin, aztreonam or ceftriaxone, then oral cephalexin. **No quinolone.**	10–14

- Give IV for high fever, leucocytosis, vomiting, dehydration, sepsis, complicated infection or outpatient failure. If blood cultures are positive, continue IV until stable; if negative, give 2–3 days IV and then oral to complete 10–14 days.
- **Obstruction** needs urgent decompression by **stent or percutaneous nephrostomy**. An obstructed solitary kidney causes acute renal failure; dose drugs by glomerular filtration rate (GFR).

ALGORITHM

1. If there is no sepsis or vomiting, then send a culture, with or without US, and treat as an outpatient with oral drugs for 7–10 d.
2. If there is sepsis, then take blood and urine cultures, image, and admit for IV treatment.
3. If improved by **72 h**, then step down to oral treatment.
4. If not improved, then repeat cultures and get a **CT**, and drain any obstruction or abscess.

KEY POINT

Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal bacterial nephritis (AFBN)

- AFBN is a lobar leucocyte infiltrate, a stage between pyelonephritis and abscess; half the patients are diabetic and half bacteraemic. On CT it is wedge-shaped and poorly marginated, with no wall or liquefaction, and seen only after contrast. An abscess, by contrast, is round with a liquid centre and a **ring sign**.
- Give IV antibiotics for ≥7 days, then 7 days oral. If it does not respond, exclude abscess, obstruction, renal cell carcinoma (RCC) and renal vein thrombosis.

Emphysematous pyelonephritis (EPN)

- P1 P3 EPN. Prognostic factors & conserv. mgt.; Prognostic indicators in emphysematous pyelonephritis — *definition, CO_2 mechanism, CT choice, Wan patterns, Huang–Tseng risk factors, drainage-first management*

Not in the paper answer:

- EPN is mostly seen in **diabetic women**. The triad is fever, vomiting and flank pain, and urine culture is almost always positive.
- Non-diabetics are usually obstructed, with less extensive disease.
- A plain film shows mottled gas over the kidney or an upper-pole gas crescent.
- Assess contralateral function with a nuclear scan.

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	50–60% mortality
Fluid with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics and percutaneous catheter drainage (PCD) or stent
2	Parenchyma, no extrarenal extension	Antibiotics and PCD
3A	Perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 or PCD fails
3B	Pararenal space	
4	Bilateral or solitary kidney	PCD (nephron-sparing)

† The risk factors are thrombocytopenia, acute kidney injury (AKI), altered consciousness and shock. Emphysematous pyelitis is often non-diabetic and milder.

KEY POINT

Fluid is a good sign and streaky gas a bad one: a contained collection can be drained; diffuse gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal and perinephric abscess

- P1 Renal and perinephric abscess — *haematogenous vs ascending, perinephric causes, insidious course, CT, empirical antibiotics, size-based drainage (≤3 / 3–5 / >5 cm), nephrectomy*

Not in the paper answer:

- Before antibiotics, abscesses were haematogenous staphylococcal; normal kidneys resist blood-borne Gram-negatives. Other risks: IV drug use, HIV, steroids, pregnancy.
- Look for a skin source 1–8 weeks earlier.
- On US an abscess is a hypo- or anechoic lesion with through-transmission.
- A renal abscess ≤**3 cm** is treated with antibiotics; one >**5 cm** needs percutaneous drainage; drain any size if the patient is immunocompromised or not responding. Aspirate if you cannot tell an abscess from a cystic tumour.
- A perinephric abscess has a lower drainage threshold (>3 cm). Rupture through Gerota's fascia gives a psoas abscess (hip held flexed and externally rotated). Only ⅓ are diagnosed at presentation, and delay causes nearly all deaths.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** is infected hydronephrosis with suppurative destruction and near-total loss of function. Patients are very ill, and **bacteriuria may be absent** if the ureter is completely obstructed. US shows a fluid–debris level.
- Give antibiotics with **immediate drainage** (stent or nephrostomy); treat the cause once stable; nephrectomy if non-functioning.

XGP and malakoplakia

- P1 P2 Xanthogranulomatous pyelonephritis; Malakoplakia — *stone–obstruction–Proteus triad, bear-paw CT, RCC mimic, complete nephrectomy; Michaelis–Gutmann bodies, intracellular antibiotics + bethanechol/vitamin C*

Not in the paper answer:

- Renal failure is uncommon in xanthogranulomatous pyelonephritis (XGP) because the disease is unilateral.
- Urine is sterile in about a third, but tissue culture is positive.
- XGP may coexist with squamous carcinoma as well as RCC or urothelial carcinoma.
- XGP: hepatic dysfunction in up to half; lymphoma is a differential (bilateral, no stones).

	XGP	Malakoplakia
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	*Proteus, E. coli*	*E. coli*
Laterality	Unilateral	Uni-/multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics with or without nephrectomy

> **KEY POINT**
> Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics; pyonephrosis and large or obstructed abscesses need drainage; when the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · Fournier gangrene and periurethral abscess

Why this matters: the skin hides the true spread, and only early radical debridement saves lives.

- **Fournier gangrene** is necrotising fasciitis of the genitalia, perineum or perianal region. A source is found in ~**95%**; patients are usually aged 50–79, and the male:female ratio is ~10:1.

- Sources are **colorectal** (perianal abscess), **urogenital** (stricture with extravasation, instrumentation, catheter) and **dermal** (hidradenitis, balanitis). Risk factors are **DM**, obesity, alcoholism, immunosuppression, trauma, paraphimosis and surgery.
- Aerobes and anaerobes act in **synergy** and cause obliterative endarteritis and thrombosis, which leads to necrosis. Infection spreads along Buck's fascia to dartos, Colles' and then Scarpa's fascia.
- Pain and oedema out of proportion come first, then crepitus, dusky skin and gangrene. **The skin underestimates fascial extent**, and systemic toxicity out of proportion is the warning sign.
- Bloods: anaemia, raised creatinine, hyponatraemia and **hypocalcaemia** (lipase-released fatty acids chelate calcium). CT shows fascial gas best, but must not delay surgery.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥ **6** suspicious; ≥ **8** strongly predictive
Fournier's Gangrene Severity Index (FGSI) †	Temperature, heart rate, respiratory rate, Na, K, creatinine, haematocrit, WBC, bicarbonate (each 0–4)	>9 predicts mortality

ALGORITHM

1. On suspicion, resuscitate and start broad IV antibiotics (piperacillin–tazobactam; or a carbapenem or vancomycin with clindamycin or metronidazole).
2. Perform **immediate radical debridement** back to bleeding fascia and leave the wound open.
3. Take the patient back for a **re-look at 24–48 h**.
4. If the source is urethral, then add suprapubic diversion; if rectal, then a colostomy.
5. Consider hyperbaric oxygen as an adjunct, then reconstruct.

- **Orchidectomy is almost never needed**, because the testicular artery is independent of the scrotal skin supply.
- Mortality is ~20–40%, and is worse with liver failure, alcoholism, DM, age, cardiac or renal failure and coagulopathy.
- A **periurethral abscess** (from stricture or catheter) needs debridement or drainage, suprapubic diversion and antibiotics.

10 · Urosepsis †

Why this matters: an obstructed infected kidney can cause septic shock within hours.

Term (Sepsis-3)	Definition
Sepsis	Organ dysfunction from a dysregulated response to infection: rise in Sequential Organ Failure Assessment (SOFA) score ≥ **2**
Quick SOFA (qSOFA)	≥2 of: respiratory rate ≥22/min, altered mentation, systolic BP ≤100 mmHg
Septic shock	Vasopressors for mean arterial pressure (MAP) ≥65 mmHg **and** lactate >2 mmol/L despite fluids

ALGORITHM

1. Hour-1 bundle: measure lactate and take blood and urine cultures.
2. Give a broad IV antibiotic within 1 h.
3. If hypotensive or lactate ≥4, then give 30 mL/kg crystalloid.
4. If still hypotensive, then start noradrenaline for MAP ≥65.
5. Achieve **source control**: decompress with a stent or nephrostomy; no definitive stone surgery now.
6. De-escalate antibiotics by culture.

† Not in the source notes. Manipulating infected stones releases endotoxin, so drain first and operate later.

11 · Special populations

Why this matters: each of these groups changes the thresholds for testing and treating.

Pregnancy

- P1 P3 Bacteriuria in pregnancy; Baceriuria and use of antibiotic policy in pregnancy — *prevalence unchanged but ascends, culture screen at 12–16 weeks, 4–7 day course, safe/avoid drug table, pyelonephritis and suppression*

Not in the paper answer:

- GFR rises by 30–50%, so creatinine falls from 0.7 to **0.5 mg/dL**; a "normal" 0.8–0.9 may mean impairment. Proteinuria is abnormal above 300 mg/24 h. The kidney grows by 1 cm.
- Aztreonam is also safe. Avoid chloramphenicol ("grey baby" syndrome).
- Severe or MDR pyelonephritis needs a carbapenem.

Elderly

- **Do not screen for or treat ASB**. Bacteriuria with confusion is not the same as UTI: without urinary symptoms, hydrate and reassess. Do not culture for non-specific symptoms or to prove clearance.
- Pyuria is common, but its **absence excludes** bacteriuria.
- The **revised McGeer criteria** (for surveillance) require non-catheterised residents to have **localising** signs and ≥10^5 CFU/mL of ≤2 species. Catheterised residents may qualify on **systemic** signs with ≥10^5.
- Treat for 3–6 days if uncomplicated and 10–14 if febrile, watching renal dosing and quinolone toxicity.

Catheter-associated UTI

- P1 Catheter Associated Urinary Tract Infections (CAUTI) — *definition, 5–10% per catheter-day, extra-/intraluminal routes, biofilm, prevention bundle, change catheter, treat only symptomatic*

Not in the paper answer:

- Catheters in place >30 days carry 4–5 species.
- The "6-C" bundle is: Consider, Connect, Clean, Closed, Call (irrigate only if needed), and Culture only if indicated.
- Silicone catheters are no better than latex.
- Stop antibiotics within 48 h of resolution.

Spinal cord injury

- All UTIs in SCI are complicated; UTI is the commonest cause of fever. Symptoms are atypical: fever, spasticity, leakage, autonomic dysreflexia † and cloudy urine.
- **Clean intermittent catheterisation (CIC, Lapides 1972)** has the fewest long-term complications and an indwelling urethral catheter the most. A suprapubic catheter delays bacteriuria but does not reduce it.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	≥10^2 CFU/mL
Catheter-free men (clean voided)	≥10^4 CFU/mL

- Do not treat ASB. Afebrile UTI: 5–7 days (7–10 if recurrent). Febrile: admit, parenteral, **14 days**. Change the catheter; image if no response in 24–48 h. Re-culture after treatment only for urea-splitters.
- Prophylaxis and methenamine do not work in NGB, and prophylaxis doubles resistance. For recurrent UTI, get imaging and urodynamics (UDS), lower storage pressure and review CIC technique.
- A catheter for >**10 years** raises the risk of bladder cancer (often squamous †), so these patients need surveillance.

Diabetes, renal transplant

- In DM, do not treat ASB, but keep a low threshold to image pyelonephritis (EPN, papillary necrosis, abscess).
- After transplant, do not screen for ASB beyond 1–2 months. Treat uncomplicated UTI for 5–10 days (not 3) and pyelonephritis for 14–21 days. Stents raise UTI by ~half, so remove them within **4 weeks**. Recurrent UTI needs PVR, imaging, VCUG and UDS.

12 · Fungal UTI

Why this matters: candiduria is usually harmless, but fungal balls obstruct and need drainage.

- P1 Fungal UTI — *Candida risk factors, asymptomatic candiduria (remove catheter; treat only neutropenia, VLBW infants, pre-procedure), fluconazole 2 weeks, amphotericin for resistant species, fungus ball drainage, IDSA vs AUA pre-procedure*

Not in the paper answer:

- AUA Best Practice Statement (BPS) 2019 lists the procedures needing antifungal prophylaxis: high-pressure endoscopy (transurethral resection of bladder tumour [TURBT], transurethral resection of prostate [TURP], therapeutic ureteroscopy) and open entry into the tract.
- For fungal balls, send a preoperative fungal culture and give 5–7 days of periprocedural antifungal.
- Give a longer course when neutropenia and obstruction coexist.

13 · Perioperative antimicrobial prophylaxis

Why this matters: one well-timed dose suffices; more only adds resistance.

Principle (AUA BPS 2019)	Rule
Indication	Any procedure breaching a tissue barrier, weighing host defence and consequences
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min

Principle (AUA BPS 2019)	Rule
Duration	**Single dose**; ≤ **24 h** only for PCNL, prostheses, special cases
Agent	Local flora per antibiogram
Known infection	Sterilise urine before elective surgery (this is treatment); infected urine causes bacteraemia in ~half

- Host risk factors: age, anomaly, malnutrition, smoking, steroids, immunodeficiency, hardware, infected stones, distant infection, prolonged stay.

Wound class	Examples	Antimicrobial prophylaxis (AP)
I clean	Scrotal/inguinal, retroperitoneal lymph node dissection (RPLND), nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; anaerobic cover
IV dirty	Abscess, open trauma	Treatment, tailor to cultures

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—
Cystoscopy with biopsy/fulguration	Host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
TURP / TURBT, laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin, with or without aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
PCNL	All	1st/2nd-gen cephalosporin; aminoglycoside and metronidazole	≤24 h
Shock-wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract	All	Cefazolin	Single
Colon segment	All	Cefazolin and metronidazole / cefoxitin / ertapenem, plus oral neomycin with erythromycin or metronidazole	Single

Procedure	AP?	Agent	Duration
Artificial urinary sphincter (AUS), penile prosthesis, sacral neuromodulation	All	Aminoglycoside and cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- Aztreonam replaces aminoglycosides in renal impairment. Vancomycin is not used routinely.
- **Transrectal ultrasound (TRUS) biopsy** causes sepsis in 0.3–3.1%, mainly from quinolone-resistant *E. coli*; the strongest risk is antibiotics within 6 months. Options are augmented prophylaxis, **targeted** prophylaxis after a rectal swab, or fosfomycin. AUA: use alternatives if local resistance is >20%. EAU: povidone-iodine rectal preparation.
- **UDS (SUFU 2017)**: give AP for neurogenic lower urinary tract dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter or recent joint implant. DM, prior UTI, valve disease or obesity alone do not need it.
- **Catheter or stent removal**: give a single dose if risk factors are present, and none for chronic catheter changes or long-term CIC. Culture the stent in high-risk patients.
- **Endocarditis** (American Heart Association [AHA]): no AP solely to prevent it. With a high-risk valve and colonised urine, sterilise the urine first (amoxicillin for enterococcus).
- **Orthopaedic hardware**: pins, plates and screws never need AP. A joint replaced <2 years ago, or a high-risk host, undergoing a high-bacteraemia procedure (stone manipulation, upper tract endoscopy, bowel surgery, transrectal biopsy) needs AP (quinolone, or ampicillin and gentamicin).

KEY POINT

Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit; otherwise give one well-timed, flora-matched dose. Anything longer is treatment.

14 · Landmark trials

Why this matters: these trials underpin current rules on prevention, ASB and male treatment length.

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen first-line after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment increased recurrence and resistant *E. coli*	Do not treat ASB; *E. faecalis* may protect
ALTAR (methenamine)	Methenamine vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative

Trial	Design	Result	Take-home
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

1 · Prostatitis: definitions and classification

Why this matters: only categories I and II are infections; most men have category III pain.

- **Drach (1978)**, built on **Meares–Stamey (1968)** segmented cultures, had four types: acute bacterial, chronic bacterial, nonbacterial, prostatodynia. MCQ National Institutes of Health (**NIH, 1999**) replaced it.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute prostatic urinary tract infection (UTI), often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI, **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretion (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory	Incidental on biopsy or semen analysis

- I and II are **5–10%**; III is **90–95%**. **IIIA vs IIIB has no clinical significance**; VB3/EPS leucocyte microscopy is no longer advised. The prostate may not be the pain source, hence "CPPS".
- **Category III (NIH Chronic Prostatitis Collaborative Research Network [CPCRN])** needs pelvic pain for **≥3 of the previous 6 months**, excluding uropathogens, urogenital cancer, prior radiation/chemotherapy, stricture and neurological bladder disease. The Multidisciplinary Approach to Pelvic Pain (MAPP) term is urologic chronic pelvic pain syndrome.

Histopathology

- Histological inflammation is near-universal (**>70%** at autopsy), so it cannot define disease.
- **Nickel grading** records location (glandular; periglandular ≤50 µm; stromal >50 µm), extent (focal <10%, multifocal 10–50%, diffuse >50%) and grade (1 separate cells; 2 confluent without destruction; 3 confluent with destruction).
- **Prostatic calculi** are a bacterial nidus. **Granulomatous prostatitis** (Epstein–Hutchins: specific, non-specific, post-transurethral resection of prostate [TURP], allergic) classically follows **intravesical bacillus Calmette–Guérin (BCG)**, giving a **transient benign prostate-specific antigen (PSA) rise**; also tuberculosis (TB).

2 · Category I: acute bacterial prostatitis

Why this matters: ABP causes sepsis and retention; ask about recent biopsy and look for abscess.

- Spread is mainly **ascending urethral**; also intraprostatic reflux, biopsy, and rarely lymphatic or blood. Risks: insertive anal intercourse, phimosis, catheters (up to ⅓ with clean

intermittent catheterisation [CIC]), instrumentation/biopsy, obstruction, preceding cystitis or epididymo-orchitis.

- ***E. coli* causes 65–80%**; also *Pseudomonas*, *Proteus*, *Klebsiella*, *Serratia*, *Enterococcus* ≤10%, and gonococcus in young men. After instrumentation, *Pseudomonas* is ~20% and abscess commoner. After biopsy, organisms are fluoroquinolone (FQ)-resistant or extended-spectrum β-lactamase (ESBL) producers, often bacteraemic. In human immunodeficiency virus (HIV): *S. aureus*, *Salmonella*, mycobacteria.

> **KEY POINT**
> **Why young healthy men get ABP:** the organism, not the host — virulent uropathogenic *E. coli* (UPEC) that forms **biofilm** more often than cystitis/pyelonephritis strains; biofilm blocks antibiotics, so prostatic infection needs longer, well-penetrating therapy.

- Symptoms are frequency, dysuria, perineal pain, fever and rigors, with possible retention or sepsis. Post-instrumentation cases are older and more septic.
- In febrile male UTI, **>90%** show raised PSA or prostate volume, so the prostate is usually involved.

Evaluation

- First ask: **spontaneous, or after manipulation or biopsy?** This decides empirical therapy.
- On digital rectal examination (DRE) the prostate is tender and boggy; be **gentle, and never massage**.
- Send blood count, midstream culture (positive in 60–85%), **blood cultures** if febrile, and renal function. Nucleic acid amplification tests (NAAT) only if discharge. Measure post-void residual (PVR) by **ultrasound (US)**, not by catheter.
- **No PSA** acutely; if raised, follow to normal (up to **6 months**). Use transrectal ultrasound (TRUS) or computed tomography (CT) only for suspected abscess; **no biopsy**.

Treatment

- About 10% progress to category II and ~10% to III. **Outpatient care only if** not systemically ill, tolerating orals and not in retention.
- Systemic illness, European Association of Urology (**EAU**): amoxicillin or a 2nd-generation cephalosporin with an aminoglycoside, or a 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%** and fully oral outpatient care, or β-lactam anaphylaxis.
- **Febrile prostatitis after biopsy** (American Urological Association [AUA] White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; use a **carbapenem**, amikacin or 2nd/3rd-generation cephalosporin.
- Oral step-down: ciprofloxacin 500 mg twice daily or levofloxacin 500 mg daily if susceptible; re-culture at 1 week.
- **Duration**: traditionally 4 weeks, but **2 weeks** ciprofloxacin is equivalent; TB prostatitis ≥ **6 months**. Non-steroidal anti-inflammatory drugs (NSAIDs) and α-blockers are useful adjuncts.
- **Retention**: suprapubic catheter is traditional; brief urethral or intermittent catheterisation is acceptable, but use **suprapubic for long-term** drainage.

Prostatic abscess

- Suspect it with high fever, immunosuppression or **no response after 48 h**. Risks: catheters, biopsy, obstruction, CBP, diabetes, cirrhosis, HIV, dialysis. Hypoechoic on TRUS; **CT** for spread and planning.

ALGORITHM

1. If the abscess is <1–2 cm, then treat with antibiotics.
2. If it progresses and lies peripherally, then perform US-guided percutaneous drainage (recurrence 15–33%).
3. If it is large or drainage fails, then perform **transurethral unroofing**, using intra-operative US to locate it.
4. If it extends outside the prostate, then drain it openly (rare).

3 · Category II: chronic bacterial prostatitis

Why this matters: CBP causes recurrent same-organism UTI in men and needs a long penetrating course.

- The prostate harbours the same organism; the man is **well between episodes** (CP/CPPS pain is continuous).
- Organisms are Gram-negatives and *E. faecalis* (main Gram-positive); special cultures for TB and *Candida*; *Trichomonas* possible. Biofilm explains poor response.
- ***Chlamydia*** is debated (urethral contamination); men with positive VB3 and negative first-void urine (VB1) improved on 4 weeks of levofloxacin and azithromycin, some evidence of causation.

Localisation test	Specimens	Interpretation
Meares–Stamey 4-glass (previous)	VB1 urethra; VB2 bladder (midstream); EPS; VB3 post-massage	Prostatic if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current)	Midstream pre-massage and first 10 mL post-massage	Predicts 4-glass in **>95%**

- No agreed cut-off (some use a **10-fold rise**). **No semen culture**. In HIV also seek anaerobes, fungi, TB.
- A DRE nodule needs cancer work-up. Recheck haematuria 4–6 weeks after clearance. PVR **>180 mL** raises infection risk.
- Men **<45** with a first UTI have normal imaging, but check for **stricture** (up to 41%). Image for diabetes, stones, voiding or neurological problems, poor response or haematuria.

KEY POINT

Why fluoroquinolones? Prostatic entry is passive (lipid solubility, ionisation, protein binding, size). FQs are lipid-soluble **zwitterions** that concentrate in alkaline CBP fluid and cover Gram-negatives including *Pseudomonas*, Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% with FQ)
C. trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg twice daily 10 days; azithromycin is better than ciprofloxacin
T. vaginalis	Metronidazole 500 mg three times daily 14 days

- Add a macrolide (it **penetrates biofilm**). Immunosuppressed: lifelong suppression. Refractory: suppression or TURP.

4 · CP/CPPS — aetiology

Why this matters: several mechanisms interact, so single treatments fail.

An insult (infection, stress, trauma) in a susceptible man starts **neurogenic inflammation**, which other factors maintain.

Domain	Key evidence
Infection	Bacterial counts are equal in patients and controls; in mice, pain from the **CP1 strain** persists after clearance
Inflammation	Biopsy inflammation in ~⅓, not correlated with symptoms; symptoms track **cytokines** (macrophage inflammatory protein-1α [MIP-1α], interleukin-17 [IL-17])
Neurological	**Central sensitisation**; normal sensory thresholds; insula changes
Pelvic floor	Tenderness and **failure to relax**; tender in **81%**
Psychosocial	**Catastrophising and helplessness predict** pain; history of abuse
Endocrine	Low, blunted adrenocorticotropic hormone (ACTH) response

5 · CP/CPPS — clinical features and evaluation

Why this matters: it is a diagnosis of exclusion; scoring and examination show which domains to treat.

- P1 P4 Chronic Prostatitis: diagnosis, classification and management; Chronic non bacterial prostatitis — *pain sites, NIH-CPSI domains, DRE/trigger points, 2-glass test, cytology, cystoscopy, UPOINT-directed multimodal therapy*

Not in the paper answer:

- **Pain defines it**, unlike benign prostatic hyperplasia (BPH): **perineum** 63%, ejaculatory pain ~half; more sites mean more severe disease.
- The **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)** † scores 0–43 (pain 0–21, urinary 0–10, quality of life [QOL] 0–12). A **≥6 points** fall is meaningful. It lacks bladder items, which the Genitourinary Pain Index (**GUPI**) adds.
- MAPP showed that pain and urinary symptoms are separate factors; depression tracks pain.
- **Bladder pain in men**: ~¾ have painful filling and ~42% meet interstitial cystitis/bladder pain syndrome (IC/BPS) criteria *see Ch 28*. **Hunner lesions** are found in ~12% and respond to fulguration.
- About 70% flare yearly; erectile dysfunction (ED) 15–40%; anxiety and depression ~2×.
- Pelvic tenderness is **muscular, not prostatic** (47% vs 6%). Severe prostatic tenderness suggests ABP instead.
- Do cytology for irritative symptoms (carcinoma in situ mimics). **PSA is not indicated**; never blame a raised value on CP/CPPS.
- After medical failure, videourodynamics may show bladder-neck obstruction or **pseudodyssynergia** (sphincter activity during voiding *without* straining), treated by physiotherapy.
- Onset clue: stress suggests pelvic floor; trauma, neuromuscular; after UTI or sexually transmitted infection (STI), infection.
- DRE nodule: cancer work-up. Confirm haematuria at ≥3 red cells per high-power field.

- Questionnaires: NIH-CPSI or GUPI, AUA Symptom Index (AUA-SI) for voiding, Sexual Health Inventory for Men (SHIM) for ED.
- Optional imaging: CT for abdominal pain, scrotal US for testicular pain, lumbosacral magnetic resonance imaging (MRI) for radiculopathy. No routine pelvic MRI or prostate biopsy.
- Cystoscopy if low flow, high PVR, or filling pain relieved by voiding. Uroflow: a young man with poor stream may have a stricture.

UPOINT phenotyping

Monotherapy fails because each man has a different mix of domains ("snowflake"). UPOINT maps them for simultaneous treatment; domain count tracks severity (UK, International Consultation on Urologic Disease [ICUD]).

Domain	Identify by	Therapy
Urinary	Lower urinary tract symptoms (LUTS), PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, stress reduction
Organ-specific	Prostate tenderness; bladder pain	†Quercetin/pollen; bladder therapy; 5α-reductase inhibitor (5-ARI) in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, irritable bowel syndrome (IBS), fibromyalgia	Tricyclics, pregabalin, acupuncture
Tenderness	Pelvic floor trigger points	Physiotherapy, myofascial release

6 · CP/CPPS — treatment

Why this matters: no single drug works, so combine modalities matched to domains.

> **KEY POINT**
> **Governing fact:** no monotherapy has a positive definitive trial — each tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team.

Modality	Evidence / role
Antibiotics	Ciprofloxacin/levofloxacin trials negative; benefit may be anti-inflammatory (lowered IL-6, tumour necrosis factor-α [TNF-α]) or VB3-only bacteria. EAU: quinolone or tetracycline ≥ **6 weeks** only if **treatment-naive and <1 year**. **No repeated courses without a positive culture**.
α-blockers	Tamsulosin, alfuzosin trials negative. For newly diagnosed, naive men with voiding symptoms or <1 year. Best combination: **antibiotic and α-blocker**.
Anti-inflammatories	Celecoxib helps pain only; steroids negative.
5-ARI	Dutasteride (REDUCE trial): ≥6-point NIH-CPSI fall in 49% vs 37%; for older men with BPH; counsel about sexual side effects.
Neuropathic	Pregabalin missed its endpoint; amitriptyline needs ≥50 mg (underdosing gives a false failure).
Conservative	Diet; **aerobic exercise**; cognitive therapy.

Modality	Evidence / role
Acupuncture	Strongest conservative evidence: sham-controlled randomised controlled trial (RCT), 61.5% vs 38.3% responders at 32 weeks. Offer after first-line failure or if drugs are declined.
Pelvic floor physiotherapy	Trigger-point release and paradoxical relaxation (Stanford protocol); myofascial therapy beats global massage (57% vs 21%). Refer if tender; works in pseudodyssynergia; urodynamics do not predict response; use experienced therapists.
Neuromodulation	Percutaneous tibial nerve stimulation (PTNS); sacral nerve stimulation (SNS) if frequency/urgency (off-label for pain).
Surgery	No TURP or prostatectomy. **Bladder neck incision** only for proven obstruction after excluding pseudodyssynergia. Fulguration of Hunner lesions.

7 · Acute orchitis

Why this matters: torsion and tumour can masquerade as orchitis.

- **Bacterial** (age >35): Gram-negatives; also syphilis, TB, brucellosis, filariasis, candiduria.
- **Mumps** is the commonest viral cause, even after vaccination; orchitis follows parotitis 2–3 weeks after exposure. **Atrophy in ~half**, infertility rare; interferon may not prevent atrophy. Also HIV, Epstein–Barr virus (EBV), SARS-CoV-2.
- **Autoimmune**: primary (anti-sperm antibodies) or vasculitic (Behçet, polyarteritis nodosa [PAN], Henoch–Schönlein purpura [HSP]); treat with steroids, immunosuppressants, intravenous immunoglobulin (IVIG).
- Ask **pain duration** (torsion is the key differential). Send urine culture (~40% positive) and STI tests; get **scrotal US**. Treat by culture/STI; in HIV give suppression (*Salmonella*). **Follow-up is mandatory** (missed torsion or tumour). Pain settles in 1–3 days, swelling in 2–4 weeks.

8 · Acute epididymitis

Why this matters: age and sexual history predict the organism.

- Epididymal pain and swelling for **<6 weeks**; infection is **ascending** (~80% urine concordance).
- **In sexually active men <35, *C. trachomatis* and *N. gonorrhoeae* dominate**; **in men >35 and children, enteric organisms dominate**. Because >40% of men under 35 grow enterics, **screen all sexually active men for STI**.
- Pain starts in the **tail**; unilateral 96%, fever ~25%, orchitis half, abscess <10%.
- Send blood count, culture, first-void urine and STI NAAT; PVR in older men. **Doppler US** excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** (intramuscular) single dose (1 g if ≥150 kg) and doxycycline 100 mg twice daily **10 days**
STI and enteric (insertive anal sex)	Ceftriaxone 500 mg IM and levofloxacin 500 mg once daily 10 days
Enteric only	Levofloxacin 500 mg once daily 10 days

- Rest, elevation, NSAIDs; admit if febrile with leucocytosis. STI: test for HIV, treat partners. **No improvement at 72 h**: repeat US for abscess or infarction (surgery).

9 · Chronic orchitis and epididymitis

Why this matters: a chronic mass may be TB or tumour.

- Testicular pain >3 months without infection is **chronic orchialgia / scrotal content pain**.
- **Granulomatous orchitis**: TB, brucellosis, syphilis, leprosy, sarcoidosis; the **idiopathic** form is usually found after orchidectomy for suspected tumour (IgG4 may contribute). **Malakoplakia** shows **Michaelis–Gutmann bodies**.
- **Chronic epididymitis** (≥ **3 months**) has a **Nickel classification**: inflammatory (infective, post-infective, TB, **amiodarone**, idiopathic), obstructive (congenital, post-infective, post-vasectomy) and epididymalgia (normal epididymis). Culture, US, reassurance.
- **TB epididymitis** is the commonest genital site (sole site in 45%): painless swelling, **sinus**. Four drugs ≥ **6 months**; surgery **only on anti-TB drugs**. **A mass not shrinking after 3 weeks needs inguinal exploration**.

10 · Chronic scrotal content pain (CSCP)

Why this matters: a cord block predicts who benefits from denervation surgery.

Intrascrotal pain ≥ **3 months** interfering with life; no evidence-based guidelines.

- Peak 20–30 y. Post-procedural: **inguinal hernia repair** (commonest; mesh/recurrent), vasectomy (~ **1 in 1000**, consent figure), scrotal surgery. Others: varicocele, hydrocele, intermittent torsion, referred pain (ureteric stone, disc, aneurysm). Idiopathic 18–25%.
- **Pelvic floor dysfunction** is common (93% symptoms, 88% high tone). **Wallerian degeneration** (84% vs 20%) affects the **"trifecta nerve complex"**: cremaster, perivasal sheath, posterior lipomatous tissue.
- **First decide**: isolated scrotal pain or part of CP/CPPS? Check **testosterone, vitamin B12**; Chronic Orchialgia Symptom Index (COSI); **duplex US**.
- A **spermatic cord block** uses **20 mL 0.25% bupivacaine** at the **pubic tubercle**. **>50% relief** predicts success of microsurgical denervation of the spermatic cord (MDSC).
- Medical: NSAIDs, **neuropathic agents** (no chronic opioids); antibiotics rarely. Serial bupivacaine and triamcinolone blocks: ~70% sustained relief. Physiotherapy if abnormal tone on DRE. Botulinum toxin equalled placebo in an RCT; pulsed radiofrequency unproven.

ALGORITHM

1. If there is an identifiable lesion, then do directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy, **microsurgical varicocelectomy**).
2. If the pain followed vasectomy, then offer **vasectomy reversal** (82–93%; wait ≥3 months).
3. If there is no lesion and the block is positive, then perform **MDSC**.
4. If the epididymis is focally affected, then perform epididymectomy.
5. Last resort: **inguinal orchidectomy**.

- Varicocelectomy works best with >7 veins, a pain score >6 and pain for >9 months.
- **MDSC** (Devine–Schellhammer 1978) is subinguinal and microscopic: divide cord nerves, **preserve arteries and lymphatics** (prevents hydrocele), ligate internal spermatic veins. Cure 50–70%, better if several structures hurt; prior failed surgery is no barrier. Consent for **testicular atrophy**. **Targeted trifecta MDSC** (cremasteric fibres, vasal sheath, posterior fat) is equally effective and quicker (~21 vs 53 min).
- **Epididymectomy** succeeds in 50–75%, best with a focal lesion; include the vasectomy site.

- **Orchidectomy** is least reliable (0–75%); **inguinal beats scrotal** (75% vs 55%), better for nociceptive pain. Persistent pain: resect the genital branch of the genitofemoral nerve.

11 · Landmark trials

Why this matters: they show why single drugs fail and which non-drug options work.

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point fall 49% vs 37%	5-ARI for older men with BPH
MAPP myofascial physiotherapy	Feasibility vs global massage	57% vs 21% responders	Physiotherapy for tender pelvic floor
Acupuncture vs sham	Large RCT	61.5% vs 38.3% at 32 weeks	Evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label series overestimate benefit

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

1 · Definition and nomenclature

Why this matters: definitions now split bladder inflammation from a pain syndrome.

- P2 P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *AUA, ESSIC (types 1–3, A–C), NIDDK, ICD-11 definitions; IC vs BPS split; phenotypes*

Not in the paper answer:

- The American Urological Association (AUA) includes "pressure" and "discomfort" deliberately, as patients use these words.
- International Classification of Diseases, 11th revision (**ICD-11**) "chronic primary bladder pain syndrome" classes the pain itself as the disease (pain worse on filling, frequency).
- **Urological chronic pelvic pain syndrome (UCPPS)**, used by the Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network, combines interstitial cystitis/bladder pain syndrome (IC/BPS) in both sexes and chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS, men) *see Ch 27*.
- BPS has never been shown to progress to Hunner lesion (HL) disease, so the two are probably distinct disorders.
- **IC/BPS** is a mixed chronic pelvic pain syndrome. It ranges from inflammatory disease with **Hunner lesions (HLs)** to pain with many non-urological complaints and a normal bladder.
- The **AUA** defines it as bladder-related pain, pressure or discomfort with lower urinary tract symptoms (LUTS) for **>6 weeks**, with no infection or other cause. † The European Society for the Study of Interstitial Cystitis (**ESSIC, 2008**) requires **>6 months** and ≥1 urinary symptom; **type 3C** is classic HL disease (C = inflammatory infiltrate, detrusor mastocytosis, granulation tissue or fibrosis).
- The National Institutes of Health/National Institute of Diabetes and Digestive and Kidney Diseases (**NIH/NIDDK, 1987**) criteria were for research; used clinically they missed **~60%**, so definitions were broadened.
- The current **trend** is to call HL disease "IC" and non-HL disease "BPS".

2 · Epidemiology (headline)

Why this matters: IC/BPS is common, under-diagnosed in men and not progressive.

- RICE study: **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms. Mean onset is 30–40 years.
- Female:male is **5:1** in clinics but **2–3:1** in populations, so men are under-diagnosed. **Ask men labelled CP/CPPS about bladder pain**, and reconsider IC/BPS when CP/CPPS treatment fails.
- **HL prevalence** is **~20%** (3.5–56%). It rises with age (uncommon under 30, over half above 60–70), equally in men.
- The course is chronic but **not progressive**, with flares (commonest trigger sexual activity; also diet, abdominal exercise). About ⅓ remit for ≥3 months.
- Childhood adversity is a risk factor. Catastrophising and depression predict suicidal ideation, so offer cognitive behavioural therapy (CBT).

KEY POINT
Uncontrolled series "work" but randomised controlled trials (RCTs) fail: regression to the mean, placebo, extra attention. Expect improvement in the first 3–4 weeks of any treatment.

3 · Aetiology and pathophysiology

Why this matters: each theory points to a treatment.

The cause is multifactorial; mechanisms form a self-perpetuating cycle.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Urinary solutes (potassium, "toxic cations") reach suburothelial nerves and cause pain and inflammation	Pentosan polysulfate (PPS), heparin, hyaluronic acid, chondroitin; potassium test
Mast cells	Histamine release causes pain and fibrosis. Marked in HL disease, sparse in BPS. Counts have **no diagnostic role**	Hydroxyzine, cimetidine, amitriptyline
Neurogenic	Neurogenic inflammation, pelvic organ cross-sensitisation, central sensitisation, hypothalamic–pituitary–adrenal (HPA) axis dysregulation; small-fibre polyneuropathy in some	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Large relative excess of systemic lupus erythematosus (SLE), Sjögren syndrome (SS), rheumatoid arthritis (RA), inflammatory bowel disease (IBD); lymphoplasmacytic infiltrate in HL	Ciclosporin A (CyA) for HL disease
Infection	No consistent organism, even by microbiome. 18-week antibiotic RCT: no benefit, more adverse events	No antibiotics without proven infection

KEY POINT
Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · Pathology: Hunner lesion vs non-Hunner phenotype

Why this matters: HL disease responds to local treatment; non-HL often does not.

- There is **no pathognomonic histology**; biopsy is normal in ~half. Biopsy only HLs or suspicious lesions to exclude carcinoma in situ (CIS), cancer, eosinophilic, tuberculous (TB) and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythema, radiating vessels, central scar; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation, dense **lymphoplasmacytic** infiltrate, increased mast cells; denudation beyond the lesion	Near-normal; sparse inflammation
Pain source	Bladder — end-organ disease	Often regional/widespread

Feature	HL disease ("IC")	Non-HL ("BPS")
Treatment	Fulguration / triamcinolone; CyA; dimethyl sulfoxide (DMSO) works better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · Associated conditions

Why this matters: pain outside the pelvis predicts failure of bladder-directed treatment.

- **75–80%** have pain beyond the pelvis and ~38% have widespread pain. True somatisation is rare.
- Among **chronic overlapping pain conditions (COPCs)**, **irritable bowel syndrome (IBS)** is commonest (>30%). Others: **fibromyalgia** (prototype nociplastic pain), chronic fatigue, migraine, temporomandibular disorder, low back pain.
- **Regional**: **high-tone pelvic floor dysfunction (HTPFD)** in **~80%**; vulvodynia; endometriosis.
- **Autoimmune** disease shows the largest relative excess: IBD (~100-fold), SLE (tens-fold), SS and RA (10–15-fold). **Trap**: in SLE think of **lupus cystitis** (fibrotic bladder with hydroureteronephrosis).

> **KEY POINT**
> Commonest is not the same as most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Widespread pain predicts failure of bladder-directed therapy.

6 · Clinical features

Why this matters: pain linked to filling is what separates IC/BPS from overactive bladder.

- Suprapubic pain is **worse with filling and relieved by voiding**, driving day and night frequency. Dyspareunia is common; isolated urethral pain uncommon.

Feature	Overactive bladder (OAB)	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting, pain-driven	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency incontinence	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive

7 · Diagnosis — exclusion plus localisation

Why this matters: missed pain generators are a common cause of treatment failure.

Basic evaluation (AUA 2022)

- **History**: pain–filling link, triggers, sexual pain, bowel, pelvic surgery or irradiation, **cyclophosphamide, ketamine**.
- **Examination**: women, diverticulum, atrophy, lichen sclerosus, mesh, **Q-tip test** (vulvodynia); men, genitals and digital rectal examination (DRE); **pelvic-floor palpation**; neurology.
- **Tests**: post-void residual (PVR), urinalysis and culture; cytology if at risk (smoking, haematuria, age >40). On a 24-h frequency–volume chart, large voids suggest a non-bladder source.

- **Scores**: O'Leary–Sant Interstitial Cystitis Symptom and Problem Indices (ICSI/ICPI; trial endpoint ≥30% fall), Genitourinary Pain Index (GUPI), Bladder Pain/IC Symptom Score (BPIC-SS), visual analogue scale (VAS) and global response assessment (GRA). † The Pelvic Pain and Urgency/Frequency (PUF) scale screens but does not diagnose. Score pain and urinary symptoms separately.

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynae, colon	Cystoscopy with or without biopsy; prostate-specific antigen (PSA); pelvic exam
Infection	Bacterial, atypical, **TB**, herpes; prostatitis	Culture; **sterile pyuria needs TB culture**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine causes upper-tract change
Outflow	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, exam, laparoscopy, colonoscopy
Neuromuscular	Pudendal entrapment; pelvic-floor myalgia	Nerve block; pelvic-floor exam

Ketamine cystitis

- Ketamine (an NMDA-receptor antagonist) causes HL-like inflammation, then fibrosis, **hydronephrosis (~30%)** and renal insufficiency (~10%). Suspect it in the **young (<30) with visible inflammation**; treat by **cessation**.

High-tone pelvic floor dysfunction

- Features: pelvic pressure, persistent urge, obstructive voiding, constipation, next-day dyspareunia; tender hypertonic muscle with **myofascial trigger points**.
- Examine six sites (each side: anterior levator, posterior levator, obturator internus) vaginally or rectally. Missed HTPFD causes poor response; refer to a pelvic-floor physiotherapist.

Urodynamics

- Not standard (AUA); use for failed therapy, incontinence, high PVR, obstruction. Expect reduced capacity, hypersensitivity and **normal compliance** (low in fibrotic end-stage). Detrusor overactivity in 15–19% **does not exclude IC/BPS**.

Office cystoscopy

- **AUA 2022**: when **HL is suspected, especially over 50**; also haematuria, failed therapy, suspected stone or cancer.
- Flexible cystoscopy shows most HLs **without hydrodistension**. Biopsy CIS-like lesions and **any trigonal lesion**. Narrow-band imaging may help.

Anaesthetic challenge vs potassium test

- **Anaesthetic challenge** (15 mL 0.5% bupivacaine with 15 mL 2% lidocaine): relief shows bladder-centred pain and selects intravesical therapy; no relief does not exclude.
- The **potassium sensitivity test** uses **0.4 M KCl** vs water. **Not recommended**: painful; negative in ¼ of NIDDK patients; positive in ~⅓ of normal men, ¼ of OAB and most CP/CPPS;

does not predict PPS response.

Cystoscopy with hydrodistension (CHD)

- Under anaesthesia, fill at **80 cm H_2O for 1–3 min**, drain and refill to see **glomerulations** and tears near the bladder neck.
- **Glomerulations are non-specific** (radiotherapy, carcinoma, chemotherapy, dialysis or diverted bladders, normal overdistended bladders) and do not match severity.
- Biopsy only **after** distension (perforation risk) and only suspicious lesions, because CIS has been misdiagnosed as IC/BPS.

8 · Phenotype-directed management

Why this matters: matching treatment to phenotype turns "failed" drugs into useful ones.

- P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *four-phenotype table: HL, anaesthetic-responsive, pelvic floor, widespread; matched treatment*

Not in the paper answer:

- Each phenotype has its own pain centre: the bladder in HL and anaesthetic-responsive disease, the pelvic floor in HTPFD, and the central nervous system in widespread pain, which needs multidisciplinary care.

> **KEY POINT**
> MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain**; amitriptyline helped **widespread pain**. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · Treatment — AUA 2022 principles

Why this matters: the aim is coping and quality of life, not cure.

- There is **no "home run"**; aim for quality of life (QoL) and coping. Set expectations; agree a flare plan.
- **No tiers**, only grouped options: patients may move to any category **except major surgery** without failing "lower" ones; severe symptoms may justify early multimodal therapy.
- Stop ineffective treatment; if not improving, seek other pain generators.
- The **Standard** is manual (myofascial) physiotherapy for pelvic-floor tenderness. Fulguration and/or triamcinolone for HLs is a **Recommendation**. Most drugs are **Options**.
- Add CBT, pudendal block and trigger-point injections (anaesthetic or onabotulinumtoxinA [BTX-A]). Chronic opioids are not routinely recommended.

ALGORITHM

1. If symptoms last ≥6 weeks, take a history, examine (pelvic floor, Q-tip, DRE), and check PVR, urinalysis and culture, with cytology if indicated.
2. Exclude confusable disease.
3. If HL is suspected (especially >50), then do cystoscopy. If HL is found, then fulgurate or inject triamcinolone; if refractory, give CyA.
4. If there is no HL, then match the phenotype: pelvic floor, physiotherapy; lidocaine responder, intravesical therapy; widespread pain, systemic therapy and CBT.
5. Combine behavioural, oral and intravesical treatment as needed, then BTX-A or sacral neuromodulation (SNM).
6. Only if bladder-centred and all else fails (rare), consider diversion with or without cystectomy.

10 · Treatment options (AUA 2022)

Why this matters: dose, evidence and risk guide a multimodal plan.

- P4 Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management — *options table: behavioural, physical (avoid Kegel), oral, intravesical, procedures, refractory HL, last resort*

Not in the paper answer:

- Behavioural measures also include fluid control, bladder training with urge suppression, and heat or cold.
- Hydroxyzine suits patients with allergy and sleep problems; heparin is slow to work.
- SNM is favoured over BTX-A for HTPFD or obstructive voiding.

10.1 Behavioural and pelvic-floor therapy

- **Diet triggers**: caffeine, alcohol, fizzy drinks, artificial sweeteners, spicy food, citrus/cranberry, tomato. **Elimination diet**: diary, bland diet, reintroduce one food at a time, avoid only confirmed triggers.
- Education and behavioural modification (**EBMP**, the control arm of the NIH amitriptyline RCT) improved 45% on its own.
- In the myofascial physiotherapy RCT, 10 sessions beat global massage, **59% vs 26%**. Adjuncts are reverse Kegels, heat, muscle relaxants, treating constipation and trigger-point injections.

10.2 Oral therapy

- **Amitriptyline** (anticholinergic, reuptake-blocking, H1-blocking, sedative): start **10 mg at night, adding 10 mg weekly to ~50 mg** (25–75). In the NIH RCT it gave 55% vs 45% (not significant), but **77% vs 53%** if ≥50 mg was tolerated, regardless of HL. **Contraindications**: long QT, conduction block, recent myocardial infarction, heart failure, ventricular arrhythmia; use caution with incomplete emptying.
- **Hydroxyzine** (H1 blocker): 25 mg at night, rising to 50 mg at night and 25 mg in the morning; its RCT was negative.
- **Cimetidine** (H2 blocker) 400 mg twice daily has one positive RCT.
- **PPS 100 mg three times daily**: only **3–6%** reaches urine; acts in **3–6 months**; the only US Food and Drug Administration (FDA)-approved oral IC drug. Phase IV RCT negative (≈40%

both arms); less effective in HL. Side effects (<4%): alopecia, gut upset, rash, bleeding; caution with high breast-cancer risk.

- **PPS maculopathy** is cumulative-dose related and may be irreversible or progress after stopping. **FDA (2020)**: eye history in all; baseline exam if eye disease; retinal exam for all **within 6 months**, then periodically.
- **Ciclosporin A**: for HL refractory to fulguration/triamcinolone; beat PPS in an open RCT; few non-HL respond. **1–3 mg/kg/day in two doses**; judge at 3–4 months. Toxicity 30–55%: **nephrotoxicity, hypertension**, infection, malignancy, gum hyperplasia. Monitor blood pressure, renal/liver function and level (1 month, then 3-monthly).
- Gabapentinoids have no RCTs; sildenafil and quercetin have only small studies.
- **Antibiotics**: AUA advises **none long-term**; one empirical course if never treated; prophylaxis only for proven recurrent cystitis triggering flares.

10.3 Intravesical therapy

For **bladder-centred** pain (lidocaine responders); a poor lidocaine response predicts failure of every instillation. AUA Option.

Agent	Regimen	Key points
DMSO 50%	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved on its own. Anti-inflammatory, analgesic, dissolves collagen. Better in HL; patients with detrusor overactivity do not respond. Garlic breath; initial flare; longer dwell causes pain. Increases absorption of co-instilled lidocaine, causing toxicity.
Heparin	10,000–40,000 U, 30–60 min, self-instilled	GAG mimic; benefit takes months; better with alkalinised lidocaine.
Alkalinised lidocaine	200 mg with 8.4% bicarbonate daily × 5	RCT 30% vs ~10% at day 3. Immediate effect, so useful for flares. Tachyphylaxis.
Hyaluronic acid, chondroitin, PPS	Weekly then monthly	Weak/negative trials

- Cocktails (DMSO, triamcinolone, bicarbonate, heparin, with or without lidocaine) weekly × 6 are no better than DMSO alone. Lidocaine toxicity (perioral numbness): drain the bladder.

10.4 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone, with biopsy if never biopsied. **75–90%** improve; repeat treatment is needed at a mean of 7–12 months (triamcinolone) or 11–29 months (fulguration).
- Resection risks perforation, and laser risks bowel injury from forward scatter. Treating glomerulations does not help.

ALGORITHM

1. Under general anaesthesia, map the HLs before distension causes bleeding.
2. Distend mildly and biopsy suspicious or trigonal lesions.
3. Fulgurate at **low volume** (Bugbee electrode), working from the **outer margin inwards**.
4. Inject triamcinolone 40 mg/mL (or 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally.
5. Fulgurate bleeders, empty the bladder and instil lidocaine–bupivacaine.

KEY POINT
Fulgurate from the margin inward: cautery erythema/oedema spreads peripherally, so the true edge is lost if you start centrally.

10.5 Hydrodistension

- AUA Option: **short (1–3 min), low-pressure (60–80 cm H_2O)**, instilling anaesthetic at the end. Benefit: ~½ at 1 month, ¼ at 3, ~none at 6. Capacity <600 mL responds better; post-procedure flare is common.
- An anaesthetic capacity **<400 mL** marks a bladder-centred phenotype, often HL.

10.6 OnabotulinumtoxinA and neuromodulation

- **BTX-A**: Option after failed conservative, oral and intravesical therapy; not FDA-approved for IC/BPS; acts on sensory pathways. **100 U** in 5 mL (200 U causes dysuria, retention) into the **trigone** (including or sparing it is equal). Lasts ~5–9 months; repeatable. The patient **must accept clean intermittent catheterisation (CIC)**.
- Transcutaneous nerve stimulation is an adjunct only, and percutaneous tibial nerve stimulation failed against sham. **SNM (S3)** is an Option and works best with **HTPFD or levator tenderness**; IC/BPS has the highest SNM adverse-event and explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centred pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD, CIC impossible
Main risk	Retention needing CIC	Adverse events, explantation

10.7 Major surgery — last resort

- Only for **rare (<10%), selected, bladder-centred** patients failing all else; pain may persist after cystectomy.
- Work-up: body map, pelvic-floor examination, intravesical anaesthetic trial, and treatment of other pain generators. In a **differential spinal block**, pain that persists after a spinal to T10 means surgery will fail.

Predicts success	Predicts failure
HLs (refractory)	No HLs
Small fibrotic bladder; anaesthetic capacity <200–400 mL	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no response
Painful filling, pain on scope touch	Urethral/neuropathic pain; HTPFD; widespread pain

- Options: conduit with bladder in place (~⅕ later cystectomy); cystectomy with conduit (~¾ pain-free); continent pouch (pouch pain 40–50%); supratrigonal augmentation (trigonal cuff recurrence, CIC). Prefer the **simplest diversion**; a retained bladder risks pain, pyocystis and haemorrhage.

11 · What NOT to do (AUA)

Why this matters: these treatments harm without helping.

Do not offer	Why
Long-term oral antibiotics	RCT negative; adverse events 80% vs 40%

Do not offer	Why
Intravesical bacillus Calmette–Guérin (BCG)	Ineffective; DMSO superior; † BCG sepsis
High-pressure, long hydrodistension †	No benefit; rupture, sepsis †
Long-term oral glucocorticoids	No benefit; toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening	Worsens HTPFD pain
Major surgery first / non-bladder-centred pain	Irreversible; pain persists

KEY POINT

Nothing is worse than irreversible surgery followed by continued pain — every predictor of surgical success marks bladder-generated pain.

12 · Landmark trials

Why this matters: they explain the physiotherapy Standard and phenotyping.

Trial	Design	Result	Take-home
Myofascial PT RCT (FitzGerald 2012)	10 sessions myofascial vs global massage	59% vs 26%	Guideline Standard
NIH amitriptyline RCT (Foster 2010)	EBMP with or without amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy works
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	Efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs by pain distribution	Hydroxyzine/PPS localised; amitriptyline widespread	Treat by phenotype

29 Sexually Transmitted Diseases

1 · Screening, prevention and partner notification

Why this matters: one sexually transmitted disease (STD) often hides another, and untreated partners cause reinfection.

- Chlamydia is the commonest reportable STD; syphilis is rising fastest.
- The nucleic acid amplification test (**NAAT**) is the standard screen. Men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Condoms** cut human immunodeficiency virus (HIV) acquisition by ~80%. They protect only covered skin; lambskin lets viruses through.
- **Partner notification** covers the previous **60 days** for urethritis, chlamydia and *Mycoplasma genitalium*; all partners for trichomoniasis; and the previous month for scabies or lice. Abstain for 7 days and until partners are treated.
- Test everyone with, or at risk of, an STD for HIV and syphilis.

2 · Urethritis

Why this matters: urethritis is the commonest male STD presentation, and correct first-line treatment prevents complications and spread.

- It causes discharge, dysuria or nothing. Complications: epididymitis, orchitis and prostatitis.
- **Gonococcal** urethritis makes up 5–20%. In non-gonococcal urethritis (**NGU**), *C. trachomatis* is commoner than *M. genitalium*, which is the leading cause of **persistent or recurrent** urethritis. Less often it is *Trichomonas vaginalis*, herpes simplex virus (HSV) or adenovirus; up to half have no pathogen.
- **Diagnosis** needs any one: discharge; Gram stain with **≥2 white blood cells (WBC) per oil-immersion field** (old ≥5 missed many); positive leucocyte esterase on first-void urine; or ≥10 WBC/hpf in first-void urine.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea; a negative Gram stain does not exclude it.
- **First-void urine NAAT** is preferred (more sensitive). **Culture** is for treatment failure, resistance surveillance and children; take it before other swabs.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (CDC 2021)	**Ceftriaxone 500 mg IM once** (1 g if ≥150 kg)	Gentamicin 240 mg IM and azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g and azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg bd 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months

Pathogen	First line	Alternatives / notes	Follow-up
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g, then 500 mg daily × 3, if macrolide-sensitive; or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- **Ceftriaxone monotherapy**: rising resistance led to a higher dose, and azithromycin was dropped. Cephalosporins are safe unless an IgE-mediated penicillin reaction occurred in the last 10 years (cross-reactivity <1%).
- **Gonorrhoea treatment failure** (symptoms beyond 3–5 days, or positive test of cure without re-exposure) needs culture and susceptibility.
- ***M. genitalium*** has no cell wall, so Gram stain misses it, and culture takes months. Use NAAT with a macrolide-resistance assay. Doxycycline lowers the load; the resistance result picks the second drug.
- **Trichomonas** is not in the first NGU work-up. Test partners of infected women, high-prevalence areas and empiric failures.
- No chlamydia test of cure: nucleic acid persists ~3 weeks.

ALGORITHM

1. Persistent NGU: check adherence and look for an untreated partner.
2. Send NAAT for *M. genitalium*, with or without *T. vaginalis*.
3. If negative, then think of chronic pelvic pain syndrome.

KEY POINT

Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease (PID) or infertility. Treating the man protects his partner. Chlamydia can also cause reactive arthritis.

3 · Epididymitis by age

Why this matters: the likely organism depends on age and sexual practice, and this picks the empiric regimen.

Acute means <6 weeks; chronic (≥6 weeks) is mostly tuberculosis (TB) *(see Ch 27)*. Always exclude **torsion** (ultrasound only if suspected). Apply urethritis criteria and NAAT; add urine culture if enterics are likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	*C. trachomatis*, *N. gonorrhoeae*	Ceftriaxone 500 mg IM and doxycycline 100 mg bd
Insertive anal sex	Above, plus enterics	Ceftriaxone and levofloxacin 500 mg daily
>35 years, obstruction or instrumentation	*E. coli*	Levofloxacin alone

Give 10 days, not 7. Doxycycline covers chlamydia, ceftriaxone gonorrhoea, levofloxacin enterics. Abstain until patient and partners are treated; test for other STDs and HIV.

4 · Genital ulcer disease

Why this matters: ulcers look alike and raise HIV risk, so every ulcer gets the same tests.

Herpes is commonest, then syphilis. Appearance misleads, so test **every ulcer** with syphilis serology (with or without darkfield microscopy or PCR), HSV NAAT and type-specific serology, a *Haemophilus ducreyi* test if endemic, and HIV. Biopsy atypical or non-healing ulcers (carcinoma mimics).

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (*T. pallidum*)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 MU IM once
Herpes (HSV-2 more than HSV-1)	**Painful** multiple vesicles that become shallow ulcers	Tender, bilateral	NAAT of unroofed base	Aciclovir and related drugs
Chancroid (*H. ducreyi*)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum, LGV (*C. trachomatis* L1–L3)	Small, painless, transient	Painful, matted, fistulae; †groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (*Klebsiella granulomatis*)	**Painless, beefy red**, bleeds easily	**Absent** (†pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV lies latent in the **sacral ganglia** and reactivates with trauma (including surgery), ultraviolet light, immunosuppression or fever.
- Asymptomatic shedding occurs on ~10% of days, so most transmission is from unaware carriers. HSV-2 recurs more, so type it. A negative NAAT does not exclude herpes; Tzanck smear is insensitive.
- Complications: aseptic meningitis and **urinary retention** (autonomic). HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg tds	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode. **Suppressive** therapy cuts transmission too. **Episodic** therapy must start within 1 day of lesions or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs all four: a painful ulcer, a typical picture, syphilis excluded and HSV excluded. Aspirate or drain fluctuant buboes. HIV-positive and uncircumcised men respond

less well.

- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma (SCC) mimics it**.
- P1 Lymphogranuloma venereum — *L1–L3; proctocolitis in MSM now commonest; groove sign; elephantiasis, rectal stricture; inguinal disease commoner in men; doxycycline 21 days*

Not in the paper answer:

- Every ulcer, LGV included, needs syphilis serology, HSV NAAT and HIV testing; biopsy non-healing lesions, as carcinoma mimics them.

5 · Syphilis staging and serology

Why this matters: the stage decides the penicillin regimen, and serology both diagnoses and proves cure.

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre and nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	
Early latent (<1 year)	Positive serology only	
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) normal
Neurosyphilis (any stage)	Central nervous system, eye, ear	**Aqueous penicillin G 18–24 MU/day IV 10–14 days**

- *T. pallidum* cannot be cultured, so **two serological tests** are used:
  - The **nontreponemal test (NTT)** (VDRL, RPR) gives false positives (HIV, autoimmune disease, pregnancy). Its titre falls with treatment, so it **monitors treatment**.
  - The **treponemal test (TT)** (FTA-ABS, TP-PA, EIA) usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT each time.

> ALGORITHM
> **Reverse algorithm**
> 1. Start with a TT. If positive, then do an NTT.
> 2. If the NTT is negative, then do a second, different TT.
> 3. If that is positive and there is no past treatment, then treat as late latent syphilis.

- The **Jarisch–Herxheimer** reaction (fever, myalgia within 24 h) is treponemal lysis, not allergy; give NSAIDs.
- With penicillin allergy and doubtful adherence, desensitise and give benzathine (IM only; neurosyphilis needs IV aqueous penicillin).
- A positive reverse-algorithm TT with documented past treatment needs no action unless there was re-exposure (then repeat the NTT in 2–4 weeks).
- In primary or secondary syphilis, test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.

- **Follow-up** serology is at 6 and 12 months (and 24 for latent disease). A fourfold rise means failure or reinfection: retreat weekly × 3, retest HIV, consider CSF. In HIV, serology may be atypical: use biopsy or darkfield.

6 · HPV and anogenital warts

Why this matters: some human papillomavirus (HPV) types cause cancer, so know when to biopsy and whom to vaccinate.

- HPV **6 and 11** cause ~90% of warts. HPV **16 and 18** cause >70% of cervical cancer, and also penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is a high-grade squamous intraepithelial lesion (HSIL). **Giant condyloma** (†Buschke–Löwenstein) is a verrucous carcinoma from HPV-6; it invades deeply despite bland histology.
- Diagnosis is clinical; typing is not needed. Inspect anal canal and meatus. **Biopsy if** the diagnosis is uncertain, there is no response or worsening, the patient is immunocompromised, or the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights/week for up to 16 weeks; wash off after 6–10 h. Works by immune stimulation. Weakens condoms
Podofilox 0.5% (patient)	Antimitotic; bd for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	Green-tea extract; tds, not washed off. Weakens condoms. Not in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid (TCA) 80–90% (provider)	Cryotherapy every 1–2 weeks; TCA weekly. Let TCA dry; neutralise with bicarbonate if painful
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if no response by 3 months. Observation is acceptable, as treatment does not clear HPV.

- The **vaccine** is **nonavalent** (6, 11, 16, 18, 31, 33, 45, 52, 58). Give it at **11–12 years** (from 9), catch-up to **26**, both sexes. Start <15: **2 doses**; ≥15 or immunocompromised: **3 doses**. It is only prophylactic; avoid in pregnancy or yeast allergy.

7 · Molluscum and ectoparasites

Why this matters: success depends on the right agent, treating contacts and spotting immunosuppression.

- **Molluscum** contagiosum forms **umbilicated** papules that regress in 6–12 months. Use cryotherapy, curettage or imiquimod. In HIV they are extensive (≥100) and worsen as CD4 falls, so start highly active antiretroviral therapy (HAART) first. Biopsy atypical lesions.
- **Pubic lice**: use a **permethrin 1%** rinse for 10 min, or, with resistance, malathion 0.5% or ivermectin 250 µg/kg, repeated at 7–14 days. Diagnosis needs live lice or viable eggs. Condoms do not protect. Review at 1 week.
- **Scabies** causes nocturnal itch and burrows. Itch is delayed hypersensitivity: 3–6 weeks first time, 1–3 days on re-infestation. Apply **permethrin 5%** to all skin, or give ivermectin 200 µg/kg repeated at 2 weeks.

- **Crusted** scabies (immunocompromised) needs topical treatment and repeated ivermectin, never lindane. Lindane 1% is a last resort (age ≥10 years). Scratching invites streptococcal infection and post-streptococcal glomerulonephritis.
- Treat contacts from the past month; wash bedding at 50 °C.

8 · HIV for urologists

Why this matters: HIV changes the organisms, course and drug interactions of urological disease.

- P1 Describe clinical implication of AIDS on the Genito-Urinary Tract? — *staging (AIDS = CD4 <200), diagnosis, GU manifestations by site (UTI, prostate, testis, voiding, ED, HIVAN, indinavir stones, cancers), PEP 72 h/28 days, needlestick 0.3%, circumcision 50–60%*

Not in the paper answer:

- Screen everyone aged 15–65 once.
- No post-exposure prophylaxis (PEP) after 72 h or for urine, saliva or tears. Needlestick: three-drug PEP even if source viral load is low.
- PrEP regimen: daily emtricitabine and tenofovir disoproxil fumarate (TDF), or tenofovir alafenamide (TAF) in men only.
- Syphilis raises HIV viral load and lowers CD4.
- Renal TB: acid-fast bacilli (AFB) culture of 3–6 early-morning urines; annual urine AFB for 10 years.
- The testis is a reservoir: give maintenance suppression (especially *Salmonella*).
- Indinavir stones need a full metabolic work-up.
- Start PEP within **72 h**, and only for mucosal, broken-skin or percutaneous contact with blood or genital fluid. Needlestick risk is **0.3%**.
- Circumcision cuts heterosexual acquisition by **50–60%** (inner foreskin is rich in Langerhans cells); it does not protect partners.

After sexual assault (men): ceftriaxone 500 mg once and doxycycline 7 days (add metronidazole for women). Give hepatitis B virus (HBV) vaccine, with or without hepatitis B immune globulin (HBIG) if the assailant is HBsAg-positive. Give HPV vaccine if aged 9–26 and non-occupational PEP (nPEP) within 72 h.

30 Cutaneous Diseases of the External Genitalia

1 · Approach to genital dermatoses

Why this matters: many genital diseases look alike, so a systematic look and early biopsy avoid missed cancer.

- Take a **history and examine**. Genital skin has few patterns, so mimics abound and the **biopsy threshold is low**.
- Ask about over-the-counter lubricants, spermicides and anti-itch creams. Do a **full skin survey**; extragenital lesions often give the diagnosis.
- Identify the **primary lesion**; on mucosa, papules and macules may appear as erosions.

Primary lesion	Definition	Secondary lesion	Definition
Macule	Flat discolouration	Scale / crust	Shed keratin / dried serum
Papule	Solid, raised, **≤0.5 cm**	Erosion	Loss of epidermis only, so **no scar**
Nodule / plaque	Solid >0.5 cm (plaque is superficial)	Ulcer	Loss of epidermis and dermis, so it **scars**
Vesicle / bulla	Free fluid ≤0.5 cm / >0.5 cm	Fissure	Linear dermal loss
Pustule	Pus	Atrophy / scar	Thinning / dermal damage
Wheal	Oedematous dermis; transient	Depth below the dermoepidermal junction decides scarring	

Bedside test	Use
Potassium hydroxide (KOH) prep; periodic acid–Schiff (PAS)	Dermatophyte (branching hyphae), *Candida* (yeast and pseudohyphae)
Mineral oil prep	Scabies mites, eggs
Tzanck smear	Herpes simplex virus (HSV), varicella zoster, molluscum
Wood's lamp	Erythrasma (coral-red); vitiligo (accentuated)
Serology / culture	Syphilis / *Pseudomonas*

Biopsy: a 2–3 mm punch is often diagnostic. Small shaft or scrotal lesions: office, local anaesthetic. Large or meatal lesions: theatre.

2 · Principles of therapy

Why this matters: thin, occluded genital skin absorbs drugs readily, so use the weakest effective treatment.

- Use **topical treatment first**. Urologists overprescribe systemic antibiotics (resistance, interactions, disturbed flora).
- Give **systemic antifungals** only for very extensive dermatophytosis, endemic mycoses, Majocchi granuloma, severe immunocompromise or topical failure. Avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- **Systemic steroid**: prednisone ≤3 weeks as a single morning dose, which limits hypothalamic–pituitary–adrenal (HPA) suppression. Long-term use causes osteoporosis, avascular necrosis (AVN) of the femoral head, cataract, diabetes and hypertension.

- **Topical steroids** cause atrophy, striae and telangiectasia in the thin genital dermis. The **prepuce acts as an occlusive dressing** (atrophy within 8 weeks). Rule: low potency, short course; lichen sclerosus (LS) is the exception.
- Use ointment emollients (creams sting). The topical Janus kinase (JAK) inhibitor **ruxolitinib** avoids steroid atrophy.

Physical modality	Indication	Caveat
Narrow-band ultraviolet B (UVB); 308-nm excimer	Psoriasis, vitiligo, dermatitis	Excimer is localised, with no non-melanoma skin cancer risk
Psoralen with UVA (PUVA)	Psoriasis, vitiligo, lichen planus (LP)	Causes **genital squamous cell carcinoma (SCC)**, so shield the genitals; now supplanted
Photodynamic therapy (PDT)	Large or resistant condyloma, warts in pregnancy	Off-label
CO_2 laser; cryosurgery	Condyloma, pearly penile papules; molluscum	Cryotherapy is used more

3 · Eczematous dermatoses

Why this matters: an itchy red genital rash is often eczema, but first exclude Candida, herpes, impetigo and drug reaction.

Atopic dermatitis (AD)

- AD is relapsing, flexural, intensely pruritic and ill-defined. **90%** present before 5 years, often with asthma or rhinitis.
- A **filaggrin defect** makes the barrier leaky; scratching lets in *S. aureus* superantigens. Biopsy is spongiotic; atopic history is the clue.
- Use soap substitutes, emollients, a short topical steroid, calcineurin inhibitors (tacrolimus, pimecrolimus), ruxolitinib and a sedating antihistamine. Oral antistaphylococcals gave no benefit (randomised trial). Severe disease: **dupilumab** or tralokinumab (anti-interleukin-13).

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct cytotoxicity; affects anyone; dose-driven	Type IV hypersensitivity in sensitised people
Share	~**80%**	Minority; genital ACD is rare (2.4%)
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest; piercing), neomycin, bacitracin, fragrances, preservatives, latex, topical steroids
Picture	Burning, limited to the exposed zone; within minutes to hours	Demarcated, pruritic, weeping
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier	Remove allergen, antihistamine. If severe, a **3-week** prednisone taper (a short course rebounds)

Seborrhoeic dermatitis (SD)

- SD gives pink-yellow greasy-scaled plaques in sebaceous areas (including glans, pubis), mainly in uncircumcised men; *Malassezia* is implicated. It affects up to **83%** of patients with

acquired immunodeficiency syndrome (AIDS) and is common in Parkinson disease, so **extensive SD should prompt an HIV test**.
- Topical **antifungal** (ketoconazole) is the backbone; hydrocortisone 2.5% only briefly.

Lichen simplex chronicus (lichenification; in the LS differential): †break the itch–scratch cycle with a short potent steroid and night sedation.

4 · Papulosquamous disorders

Why this matters: infection, syphilis and in-situ cancer can all look like psoriasis.

The differential includes tinea, erythrasma, secondary syphilis and neoplasia (Bowen disease, extramammary Paget disease (EPD), mycosis fungoides). "Psoriasis" failing treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- Psoriasis affects ~2% (⅓ familial). Triggers: trauma (Koebner), streptococci, stress, drugs. Onset peaks at 20–30 and 50–60 years.
- It forms sharp plaques with silvery scale; check elbows, knees, nails, scalp, **umbilicus** and natal cleft. **Inverse psoriasis** loses scale in folds, so use KOH to exclude tinea or *Candida*.
- Histology: **"sandwich sign"** (alternating neutrophils and parakeratosis).

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%) continuously for **≤2 weeks**; triamcinolone 0.1% for thick plaque
Steroid-sparing	Calcipotriene, calcineurin inhibitors, low-potency retinoids
Localised	308-nm excimer laser
Contraindicated	**PUVA** (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, retinoids, apremilast, anti-tumour necrosis factor (anti-TNF), ustekinumab, **ixekizumab (anti-IL-17), which has the best genital data**

Reactive arthritis (formerly Reiter)

- Reactive arthritis follows urethritis (*Chlamydia*, gonococcus, *Ureaplasma*) or enteric infection in **HLA-B27**-positive people. It causes conjunctivitis, arthritis and oral ulcers (⅓ complete).
- **Balanitis circinata** and **keratoderma blennorrhagicum** look and stain like psoriasis, so diagnose from **context**. It is self-limiting; use low-potency steroid.

Lichen planus (LP) and lichen nitidus

- LP forms **polygonal, violaceous, flat-topped papules**. Glans papules are white, reticular or annular; Koebner occurs. Look for drug or contact triggers.
- Papular LP resolves in ⅔ by 1 year. **Erosive mucosal LP persists, scars and carries an SCC risk**, so review regularly and biopsy ulcers.
- Use an **ultrapotent** steroid (clobetasol 0.05%), with or without a calcineurin inhibitor; systemic steroid if severe.
- **Lichen nitidus**: tiny clustered flesh-coloured papules; resolves within a year.

5 · Lichen sclerosus (LS) / BXO

Why this matters: LS scars the foreskin, meatus and urethra and can become cancer.

- P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — *scarring disease; phimosis, then meatus, then anterior stricture; histology; HPV-independent SCC 2.3–9%; clobetasol 3 months, then maintenance; circumcision; BMG not genital skin*

Not in the paper answer:

- Histology also shows follicular plugging.
- Circumcision is usually curative, but LS may recur in the scar.
- Steroid evidence is strongest in women and children.
- LS is 6–10× commoner in females; males are usually >60. **15–20%** have extragenital disease. In men the **perianal skin is spared**, and purpura may mask the whiteness.
- The differential includes erosive LP, lichen simplex, morphoea, pemphigoid and child abuse.

ALGORITHM

1. Biopsy.
2. Give a **potent steroid (clobetasol 0.05%) for 3 months**, then taper.
3. Continue twice-weekly maintenance (or mometasone 0.1% or a calcineurin inhibitor).
4. Lifelong surveillance of non-healing fissures, ulcers, nodules.

- Recurrence is **40–50%**, so maintain proactively. Calcineurin inhibitors are inferior; refractory disease gets retinoids or PDT. †Meatal disease needs meatotomy or meatoplasty.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months, then maintenance

KEY POINT

Steroid strength tracks the scarring and malignant stakes of the disease. LS is the single exception to the short-course rule.

6 · Vesiculobullous and drug eruptions

Why this matters: a genital drug rash may be trivial or life-threatening Stevens–Johnson syndrome (SJS).

Feature	Erythema multiforme (EM) minor	SJS (EM major)	Fixed drug eruption (FDE)
Trigger	**HSV** (10–14 days before)	**Drugs**: sulfonamides, NSAIDs, penicillins, cephalosporins, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, barbiturates, carbamazepine; rarely finasteride, tadalafil, fluconazole
Lesion	Symmetric **targets**; palms, soles, mouth	Prodrome, then macules, blisters and necrosis; genitals involved in ~**60%**	Solitary violaceous plaque with a halo, often on the glans
Course	Self-limited, recurs	4–6 weeks; mortality up to ~**30%** (overlap with toxic epidermal necrolysis, TEN)	**Recurs at the same site within 24 h** of re-exposure
Sequelae	None	Meatal stenosis, synechiae, strictures	Brown pigmentation

Feature	Erythema multiforme (EM) minor	SJS (EM major)	Fixed drug eruption (FDE)
Treatment	Antihistamine; **suppressive aciclovir prevents it**	See algorithm	Stop the drug

ALGORITHM

SJS

1. Stop the offending drug.
2. Give burn-type supportive care in an intensive care or burn unit (lowers mortality), with multispecialty wound care.
3. Limited data support cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis or intravenous immunoglobulin (IVIG); systemic steroids are controversial.

KEY POINT

One drug class can cause a trivial FDE or a lethal SJS; extent and systemic illness separate them. Genital FDE mimics herpes; "same spot each time" is the clue.

† **Autoimmune blistering**: pemphigus vulgaris is intraepidermal (flaccid blisters, Nikolsky positive); pemphigoid is subepidermal (tense); mucous membrane pemphigoid scars (meatal stenosis). Use perilesional biopsy and direct immunofluorescence.

7 · Noninfectious genital ulcers

Why this matters: these are diagnoses of exclusion; a genital ulcer is infectious until proven otherwise (see Ch 29).

Behçet disease (BD)

- BD is a relapsing **HLA-B51**-linked vasculitis, commonest in Turkey and **more severe in males**. It causes oral and genital ulcers, uveitis, epididymitis, thrombophlebitis and **pulmonary artery aneurysm**.
- Oral ulcer is most sensitive; **genital ulcer is most specific**. No confirmatory test exists.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Treatment: colchicine, dapsone, steroids, cyclosporine or **anti-TNF**.

Pyoderma gangrenosum (PG) and factitial

- PG occurs with inflammatory bowel disease, arthritis, myeloproliferative disease and HIV: a painful ulcer with an **undermined violaceous border** and pathergy. Biopsy the edge (neutrophils). Treat with systemic and local steroids, with or without cyclosporine, anti-TNF or IVIG. †Avoid debridement.
- **Factitial dermatitis**: oddly angulated erosions; psychiatric association.

8 · Infections

Why this matters: the depth of infection decides how urgent treatment is.

Balanitis and balanoposthitis

- Children: bacterial. Adults: hygiene, trauma, irritants, *Candida*, bacteria or viruses.

ALGORITHM

1. Remove irritants and improve hygiene.
2. Give a topical antibiotic or antifungal, then a short low-potency steroid.
3. If this fails, then **biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV).
4. If recurrent with phimosis, then **circumcision**.

Type	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny moist red plaque, elderly uncircumcised	Biopsy, then circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Mica-like scaly plaque in men circumcised as adults	Biopsy; excision

Soft-tissue infection by depth

Entity	Level / organism	Sign	Management
Folliculitis	Follicle; *S. aureus*; *Pseudomonas* after a hot tub	Perifollicular pustules	Hygiene; culture if refractory
Furuncle	Deep follicle abscess; *S. aureus*	Tender boil	Incision and drainage (I&D); cover methicillin-resistant *S. aureus* (MRSA); nasal mupirocin if recurrent
Erysipelas	Upper dermis and lymphatics; *S. pyogenes*	Sharp border	Antibiotics
Cellulitis	Deep dermis and subcutis; mixed in diabetes	Indistinct border	Systemic antibiotics, with or without MRSA cover; **mark the border**
Fournier gangrene	Fascia; polymicrobial	Pain out of proportion, grey skin, crepitus	Emergency debridement *(see Ch 26)*

KEY POINT

Fournier gangrene is fascial, so skin shows a fraction of the necrosis; testes are spared (own blood supply). Imaging (CT is best for gas) must never delay surgery. Debride to bleeding tissue, with a second look at 24–48 h.

- **Ecthyma gangrenosum**: *P. aeruginosa* bacteraemia in the immunocompromised; a haemorrhagic bulla becomes black eschar. Give immediate IV antipseudomonals, with or without debridement.
- **Genital bite** (*Eikenella corrodens*): amoxicillin–clavulanate 875/125 mg bd, with irrigation and debridement.

Hidradenitis suppurativa (HS)

- HS affects apocrine-bearing skin after puberty, more in females and Black ancestry; up to 41% familial. It is **follicle-centred**: plugging and rupture lead to abscesses, **sinus tracts** and scars.
- Complications: fistulae to urethra, bladder or rectum, lymphoedema, amyloidosis and **SCC in chronic scars**.

† Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, deroofing
II	Recurrent abscesses, tracts, scarring; separated areas	Oral tetracycline, or clindamycin and rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts	Adalimumab and **wide excision**, graft or flap

- Stop smoking, lose weight. Topical clindamycin equals oral tetracycline (randomised trial). Others: dapsone, acitretin, metformin, spironolactone (women), 5α-reductase inhibitors (men) and **adalimumab 40 mg weekly**. **Avoid lithium**.
- **Repeated incision and drainage is discouraged** (the tract recurs); chronic disease needs wide excision with grafting, or CO_2 laser.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Organism	*Candida*	Dermatophytes	*Corynebacterium minutissimum*	None
Scrotum	**Involved**	**Spared**	—	May be involved
Look	Macerated; **satellite lesions**	Scaly advancing border, central clearing	Red-brown scaly patch	Non-scaling plaque
Test	KOH: yeast and pseudohyphae	KOH: hyphae	Wood's lamp: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; occasional fluconazole 150 mg; test for diabetes	Topical azole or allylamine; deep infection: terbinafine 250 mg/day for 1 week; no steroids	Antibacterial soap, clindamycin 1%, miconazole; oral erythromycin if severe	Low-potency steroid

KEY POINT

Azoles cover yeast and dermatophyte, so use them when unsure. Nystatin misses tinea; allylamines are weaker on *Candida*. A negative KOH from an untreated lesion means rethink the diagnosis.

Infestations

Feature	Pediculosis pubis	Scabies
Sign	Lice/nits on hair; eyelashes	Nocturnal itch; **serpiginous burrows**; crusted in the immunosuppressed
First-line	Permethrin 5% to hair-bearing areas, **repeat at 1 week**	Permethrin 5% to the whole body overnight, **repeat at 1 week**, treat the household

Feature	Pediculosis pubis	Scabies
Alternatives	Oral ivermectin; malathion 0.5%	Ivermectin 200 µg/kg ×2; lindane disfavoured (central nervous system toxicity)
Extra	Screen for gonorrhoea/chlamydia	Postscabies nodules on the glans: intralesional triamcinolone

9 · Premalignant and malignant lesions

Why this matters: in-situ penile cancer is curable with organ-sparing treatment if biopsied early.

- P1 Premalignant conditions in urology (including premalignant lesions of penis) — *PeIN HPV vs non-HPV; Queyrat, Bowen, bowenoid papulosis; Buschke–Löwenstein (no RT); glans therapy (5-FU, imiquimod, laser, resurfacing)*

Not in the paper answer:

- PeIN is HPV-related (types 16, 8) or non-HPV (LS, chronic LP); thermal injury is another risk.
- Bowen disease: solitary, sharp red scaly plaque (verrucoid on the shaft); indolent.
- Bowenoid papulosis shows discontinuous atypia.
- Verrucous carcinoma: local excision or Mohs surgery; no primary radiotherapy (anaplastic transformation).
- Bowenoid papulosis: tazarotene or electrocoagulation also work.
- Other risks: immunosuppression, radiation, arsenic. Bowen disease presents in the seventh decade, bowenoid papulosis in the third (it **regresses**; female partners need cervical screening). Biopsy **several areas**.

Lesion	Treatment
SCC in situ (SCCis) — accessible (scrotum)	Excision with a **5-mm margin** (favoured)
SCCis — glans	Mohs surgery, laser, cryotherapy; topical 5-fluorouracil (5-FU) or imiquimod (higher recurrence)
Bowenoid papulosis	Observe in a reliable young man; or 5-FU 0.5%, imiquimod, cryotherapy, laser; relapse is common
PEKMB	Biopsy (exclude verrucous carcinoma/SCC); excision or ablation, 5-FU; glans resurfacing if refractory

KEY POINT

HPV mapping: types 6 and 11 cause condyloma and verrucous carcinoma (locally destructive, rarely metastasises); type 16 causes bowenoid papulosis and SCCis.

Tumour	Key features	Treatment
Basal cell carcinoma	Rare; pearly papule with telangiectasia	Excision or Mohs; advanced: vismodegib, sonidegib
Kaposi sarcoma	Human herpesvirus 8 (HHV-8); classic, AIDS (genital in ~3%) and transplant types; a glans lesion may block the meatus	Restore immunity (antiretrovirals; swap calcineurin for mTOR inhibitor). Solitary: excision, laser or intralesional vinblastine. Regional: radiotherapy 15–30 Gy. Disseminated: chemotherapy
Melanoma	<1% of penile cancers	Wide excision or partial penectomy; advanced: BRAF/MEK inhibitors, anti-CTLA-4, anti-PD-1

Tumour	Key features	Treatment
Extramammary Paget	Red plaque at apocrine sites; CEA and EMA positive; underlying carcinoma in **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision or Mohs; imiquimod, 5-FU, radiotherapy
Mycosis fungoides	Pruritic patches progress to plaques, then tumours; Sézary syndrome if in blood	Ultrapotent steroid, nitrogen mustard, phototherapy, electron beam

10 · Benign male genital lesions

Why this matters: normal variants mistaken for warts or cancer get needless treatment.

Lesion	Site / look	Mimic	Management
Pearly penile papules	Coronal rows; 14–48% of young men; angiofibromas	Condyloma (but no HPV)	Reassure; laser or cryotherapy if cosmetic
Fordyce spots	Pin-sized shaft papules (ectopic sebaceous glands)	Warts	Reassure
Angiokeratoma of Fordyce	1–2 mm red-purple scrotal papules; may bleed	Fabry disease	Observe; laser if bleeding
Zoon balanitis	Shiny moist red glans plaque; elderly uncircumcised men; plasma cells	Queyrat, EPD	**Biopsy**, then **circumcision (curative)**
Sclerosing lymphangitis	Coronal cord after vigorous sex	—	Self-limiting
Median raphe cyst	Ventral, near the glans; no urethral link	—	Excision

> **KEY POINT**
> Zoon balanitis and erythroplasia of Queyrat are both red glans plaques in uncircumcised men: one benign and cured by circumcision, the other SCCis. A persistent red glans plaque always needs biopsy.

11 · Pigmentary disorders

Why this matters: white genital patches raise the question of vitiligo versus LS.

- Vitiligo is accentuated by Wood's lamp; treat with narrow-band UVB or excimer. †It is not premalignant and, unlike LS, has normal texture with no atrophy or scarring.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

1 · Urogenital tuberculosis — organism and pathogenesis

Why this matters: tuberculosis (TB) seeds the urinary tract years after lung infection and heals by scarring, so it presents late and obstructs.

- P3 P2 Renal and genitourinary TB: etiopathogenesis, presentation, diagnosis and management — *MTBC, granuloma/IFN-γ, latency and reactivation, routes, renal sequence to autonephrectomy, site frequencies*

Not in the paper answer:

- TB is the second "great imitator" after syphilis; untreated, it irreversibly causes renal failure and infertility.
- Latency can last decades (46 years), so UG TB often appears without active lung disease.
- Isolated UG TB is generally **not infectious**, except by skin inoculation or aerosol-generating procedures (theatre).
- Contiguous spread runs from spine to psoas to kidney; gut TB can cause enterorenal or enterovesical fistula.
- **Urogenital (UG) TB** is preferred, because the urinary tract is involved far more than the genital tract. ***M. bovis* (and bacille Calmette–Guérin, BCG) is innately pyrazinamide-resistant**; *M. africanum* occurs in West Africa.
- UG TB is **15–20%** of pulmonary TB in developing countries. Infection stays latent in 90–95%; lifetime reactivation is **5–15%**.
- Before cellular immunity develops (up to 12 weeks), bacilli spread via hilar nodes and blood. The granuloma (epithelioid cells, Langhans giant cells and a lymphocyte cuff) is maintained by interferon-γ (**IFN-γ**), with interleukin-2 and tumour necrosis factor-α. Reactivation: old age, renal failure, diabetes, malnutrition, human immunodeficiency virus (HIV), immunosuppression.
- Ascending spread: intravesical BCG causes UG TB in **~1%**.

> **KEY POINT**
> UG TB is blood-borne and lands in the two best-perfused UG organs, kidney and epididymis. Blood-borne seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · Clinical features and pathology by site

Why this matters: symptoms are vague; sterile pyuria is the clue that should trigger a TB work-up.

- It is indolent; patients are repeatedly treated for urinary tract infection (UTI) or investigated for cancer. Constitutional symptoms occur in **< 20%**. **Lower urinary tract symptoms (LUTS) dominate**; haematuria or flank pain occurs in ~⅓ and colic in <10%. **The hallmark is sterile pyuria** (>90%).

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities, then autonephrectomy (up to ⅓); "putty" kidney; end-stage renal failure ~7%; pelvic squamous metaplasia gives squamous cell carcinoma (SCC) risk

Site	Frequency / route	Signature lesion
Ureter	Descending	Stricture at the vesicoureteric junction (VUJ); beaded corkscrew ureter; may obstruct and reflux. **Stricture obstruction is an important cause of renal failure**
Bladder	Descending	Starts at orifices; dome worst, trigone spared. After ~1 year <100 mL; "thimble" <20 mL
Epididymis	10–55%; second commonest; bilateral in ~⅓	Starts in the **globus minor (tail)**; posterior scrotal sinus in up to half; spreads to testis
Vas	Contiguous	Thickened, beaded
Testis	From epididymis; isolated disease rare	Hard tumour-like mass; hydrocele ~5%
Prostate	22–49%	Calcified hard gland. "Prostatitis" responds partly to quinolones, then relapses (quinolones are active against MTBC)
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia; infertility may be the first symptom
Urethra	1.9–4.5%	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer; **papulonecrotic tuberculid** (hypersensitivity; culture and PCR negative; diagnosed by response to treatment)

- Kidney disease without imaging change: granulomatous interstitial nephritis (reversible renal failure), glomerulonephritis, amyloidosis or miliary TB.

3 · Diagnosis

Why this matters: bacilli are shed intermittently and sparsely, so diagnosis needs repeated cultures.

- P3 Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management — *3–5 early-morning urines, ZN, LJ/MGIT culture (gold standard, monitors treatment), GeneXpert, LF-LAM, TST/IGRA, imaging signs, cystoscopy, algorithm*

Not in the paper answer:

- Urinary inhibitors lower NAAT sensitivity; TST is usually positive in UG TB.
- KUB calcification progresses from punctate to globular to ring (papillary necrosis); an arrowhead stone may sit in a hiked-up pelvis. IVU may show cloverleaf calyces.

ALGORITHM

Hierarchy of proof

1. If culture or MTBC DNA is positive, then the diagnosis is made.
2. If negative, then caseating granulomas on tissue in the right context confirm it.
3. If still negative, then a compatible picture, exposure and response to empirical treatment suffice.
4. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

- Culture sensitivity is 10–90% (intermittent shedding). The Ziehl–Neelsen (ZN) smear detects **< 50%**. Acidic urine inhibits mycobacteria: culture promptly.
- **GeneXpert** detects rifampicin resistance; >90% of such strains are also isoniazid (INH)-resistant, so it is a surrogate for multidrug-resistant (MDR) TB. The World Health Organization (WHO) uses it in the initial work-up of all TB; in the USA it is an adjunct, and a

negative nucleic acid amplification test (NAAT) does not exclude TB. It **cannot monitor treatment** (dead bacilli shed DNA).

- Biopsy fits TB in only ⅕–⅓; fine-needle aspiration cytology of an epididymal nodule is diagnostic in ~⅔. At cystoscopy ulcers mimic cancer; a **"golf-hole" orifice** means imaging the upper tract.
- Tuberculin skin test (**TST**) cut-offs: **≥ 5 mm** for contacts and the immunosuppressed; **≥ 10 mm** for immigrants, high-risk workers and drug injectors; **≥ 15 mm** with no risk factors. False negatives: 10–25%. Interferon-γ release assay (**IGRA**) has no BCG false positives and is the preferred screen for people from endemic countries. Both show exposure only.

Modality	Signs / role
Plain kidney–ureter–bladder film (KUB)	Calcification in 24–44%; **lobar "putty"/"cement" kidney** is pathognomonic of end-stage disease; ureteric calcification is **intraluminal** (schistosomiasis: mural)
Intravenous urography (IVU)	**Moth-eaten calyx**; **phantom calyx**; **Kerr's kink**; **pipe-stem ureter**; beaded corkscrew ureter; contracted bladder. Findings are mostly **obstructive**
CT urography	Standard; granuloma may mimic renal cell carcinoma; less sensitive than IVU for earliest papillary change
Ultrasound (US)	**Serial monitoring of hydronephrosis during treatment**
MRI	Diffusion-weighted imaging separates pyonephrosis from hydronephrosis and grades fibrosis; avoid gadolinium in renal failure

KEY POINT

Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). So obstruction is the commonest finding and must be watched during treatment.

4 · Medical treatment

Why this matters: drugs cure most UG TB, but healing fibrosis can worsen obstruction during treatment.

- Combine drugs, because bacilli exist in different metabolic states and combinations prevent resistance. Check baseline liver function tests (LFTs), renal function, platelets, HIV; use directly observed therapy (DOT). First-line drugs reach high urinary levels.

ALGORITHM

Standard 6-month short course (2HRZE / 4HR)

1. Intensive, 2 months: daily INH (H), rifampicin (R), pyrazinamide (Z) and ethambutol (E); stop E once fully susceptible.
2. Continuation, 4 months: HR daily (or thrice weekly by DOT; twice weekly no longer recommended) to eradicate persisters.
3. Give pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, flu-like illness, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin with antiretrovirals (uveitis)

Drug	Daily adult dose	Signature toxicity	Monitor / note
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, arthralgia, hepatitis	*M. bovis* is resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green vision)	Check vision; the **only non-hepatotoxic** first-line drug

- **Not for UG TB**: weekly rifapentine; 4-month rifapentine–moxifloxacin. Complicated disease: some give 12–24 months (or ≥6 months after surgery), as UG TB relapses more.
- **Monitor** LFTs, vision, urine culture and **serial US**: healing fibrosis can worsen strictures and bladder contraction.
- **Steroids** are proven in meningitis and pericarditis but anecdotal in UG TB; consider them if a stricture worsens on treatment. Rifampicin increases steroid metabolism†.

5 · Surgical treatment

*Why this matters: about half of patients need surgery, now mostly **reconstructive**, and timing decides whether the kidney is saved.*

- The goal is organ preservation. Indications: obstruction or abscess, a non-functioning infected kidney, resistant hypertension, reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy**, except in emergencies.

> **ALGORITHM**
> 1. If uraemia or sepsis, then decompress urgently.
> 2. Try a retrograde double-J stent first (preferred; succeeds in < half, 41%).
> 3. If that fails, then try an antegrade stent.
> 4. If that fails, then percutaneous nephrostomy (PCN); loculated systems may need several.
> 5. Then repair definitively; if the kidney is unsalvageable, then nephrectomy.

- Avoid high contrast pressure (dissemination). Removing a PCN without fixing the cause leaves a tuberculous cutaneous fistula.
- **Nephrectomy is established** for a non-functioning kidney with recalcitrant TB or resistant hypertension (improves in ~⅔). **Routine removal** of an asymptomatic non-functioning kidney is **contested**. Partial nephrectomy suits a polar lesion unresponsive to drugs.
- Technique: oblique extendable retroperitoneal incision (dense fibrosis); remove caseous perinephric tissue en bloc. **Ligate artery and vein separately** (late arteriovenous fistula). Usually leave the ureter; avoid pleura and peritoneum. Laparoscopy is increasingly preferred.
- **Strictures**: lower ureteric is commonest. Endoscopic results are poorer (ischaemic fibrosis), best if short with a residual lumen. Stent early for strictures arising on treatment; balloon dilatation often fails. **Image every stricture**; if worsening, give steroids; **if there is no improvement after 6 weeks, operate**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (lower pole)	Cover with capsule or omentum
Upper/mid ureter	Ureteroureterostomy; Davis intubated ureterotomy	—

Site / gap	Procedure	Key point
Lower ureter	Excise, then ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity; avoid genitofemoral and femoral nerves
Long gap	Boari flap, with or without hitch	**10–15 cm**; poor in a contracted bladder
Multiple / recurrent	**Ileal ureter**	Native ureter unusable

KEY POINT

Trap: in TB the bladder may itself be contracted, ruling out psoas hitch and Boari flap. Assess capacity first; ileal ureter, with or without augmentation, is then the option.

- P2 Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male — *assessment, augmentation vs substitution, ureteric reimplantation, bowel complications, follow-up*

Not in the paper answer:

- Symptoms that justify surgery: intolerable frequency, urgency, pain or haematuria.
- Contraction is commoner with late diagnosis; the spared trigone allows supratrigonal substitution.
- Urethral strictures are treated endoscopically; a fistula needs suprapubic diversion and delayed repair.
- Contracted bladder: operate for intolerable symptoms or capacity **< 100 mL**, once chemotherapy controls disease; a thimble bladder needs orthotopic substitution. Incise a bladder neck contracture.
- **Genital** surgery is only for failed medical therapy. Testis spared: **epididymectomy**, starting at the globus minor and ligating the vas at the external ring (it is diseased up to there). Testis involved: scrotal orchidectomy. Obstructive azoospermia is often irreversible, so offer sperm retrieval with assisted reproduction†.

6 · BCG-related infection, relapse, special situations

Why this matters: urologists cause BCG infection, and TB relapses late.

- **BCG UG TB** (~1%) causes granulomatous prostatitis, epididymo-orchitis, renal abscess, ureteric obstruction or sepsis; Mantoux is unhelpful. †For local cystitis, withhold BCG and give a quinolone. For organ infection, give INH and rifampicin (with or without E) for 3–6 months; **pyrazinamide is useless**. Sepsis: triple therapy and steroids; stop BCG permanently.
- **Relapse** occurs in **6.3–22%** (pulmonary 2–6%), at a mean of ~5 years. Follow up for **10 years** and culture any recurrence.

Situation	Change	Duration
MDR-TB (INH and R resistant)	Second-line; BPaL (bedaquiline, pretomanid, linezolid), not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR plus fluoroquinolone and bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	Treat promptly; HRE and pyridoxine; Z omitted in the USA (WHO includes it); breastfeeding allowed	9 months (without Z)

Situation	Change	Duration
HIV on effective antiretroviral therapy (ART)	Standard daily regimen; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or a rifamycin-free regimen	≥ 18 months if rifamycin-free

- **HIV** raises TB risk 15–20-fold; disease is more disseminated, with **less fibrosis and fewer strictures**, and higher mortality. **Transplant**: fever within 6 months with normal imaging needs a **urine acid-fast bacilli (AFB) culture**.

7 · Schistosomiasis (bilharzia)

Why this matters: trapped eggs cause obstruction and squamous bladder cancer, often after infection has burned out.

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical/pelvic plexus — UG tract	***Bulinus***
S. mansoni	Africa, Middle East, S America	Mesenteric — liver, gut	*Biomphalaria*
S. japonicum	China, SE Asia	Mesenteric — liver, gut	*Oncomelania*

ALGORITHM

Life cycle

1. Fork-tailed cercariae penetrate intact skin in fresh water, become schistosomulae and pass via the lungs to the liver, where they pair (female in the male's gynaecophoric canal).
2. Paired adults move to the vesical plexus (80–110 days) and live 3–5 years.
3. Eggs (~140 µm, **terminal spine**) cross the bladder wall into urine and hatch as miracidia.
4. Miracidia infect *Bulinus*, which releases cercariae.

- **Retained eggs cause the disease**. Adults evade immunity (tegument) and do not multiply in humans. Children aged 5–15 carry the heaviest burden; dams and irrigation spread it.
- The **egg granuloma** forms polypoid masses (active), then fibroses and calcifies into **"sandy patches"**, leading to obstruction, metaplasia and cancer. Hydroureter precedes hydronephrosis, and tubular function is lost before glomerular function.
- Obstruction is **distal** (intramural or juxtavesical); meatus ~1%. Hydroureter may be **segmental**, **tonic** (reversible once relieved) or **atonic** (fibrotic, failed).
- ***Salmonella*** shelters in the tegument, causing chronic bacteriuria or bacteraemia; treat both.
- **Bladder cancer**: the International Agency for Research on Cancer (IARC) rates it **Group 1**. Hyperplasia progresses via squamous metaplasia and dysplasia to carcinoma, driven by inflammation, VEGF and bacterial nitrosamines alkylating DNA (p53). Patients are younger (40–50); **SCC is 60–90%**, adenocarcinoma 5–15%, often posterior wall. Mass drug administration (MDA) in Egypt shifted SCC to urothelial cancer.
- **Acute**: swimmer's itch; **Katayama fever** at 2–8 weeks (rare with *S. haematobium*). **Chronic** (from 8–12 weeks): **painless terminal haematuria**.
- Stages: active (polyps), late chronic active (contracted bladder ~50 mL, trigone spared), quiescent, and chronic inactive (no viable eggs), where 40–60% with obstruction first present.
- **Female genital** disease (⅓–¾ of infected women) gives cervical and vaginal sandy patches, infertility and possibly HIV risk. **Males** get haemospermia (eggs in semen before urine),

funiculitis mistaken for tumour, and infertility (often reversible).

> **KEY POINT**
> Trap: a negative urine egg count does not exclude severe disease; the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Collect **09:00–15:00** (noon peak); negative in the first 8–12 weeks and in light or inactive disease
Rectal snip (squash prep)	Active or past infection; egg viability	Avoids bladder biopsy risks
Serology (FAST-ELISA and Western blot)	Exposure (>90%; from 4–6 weeks)	Stays positive, so it **never judges cure**
Circulating anodic/cathodic antigen	Live worms	Urine circulating cathodic antigen (CCA) is poor for *S. haematobium*
PCR	Most sensitive	Impractical in the field

- Imaging: US shows wall thickening, polyps, hydronephrosis. Plain film shows a **calcified bladder ("fetal head in the pelvis")** and mural ureteric calcification. IVU shows ureteric striations (earliest); voiding cystourethrography shows reflux in ~25%. Cystoscopy: sandy patches, polyps, ulcers, golf-hole or stenotic orifices, tumours.
- **Praziquantel (PZQ)** is the only WHO-listed drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart. Cure or egg reduction is 60–90%; pregnancy FDA category B. Add steroids for Katayama fever. PZQ is weak on schistosomulae, so repeat after several weeks. Artemisinins act on schistosomulae but are avoided where malaria is co-endemic.
- Active polypoid obstruction resolves with PZQ alone (1–2 months; children 2–6 weeks); late fibrotic stenosis (up to 80%) needs surgery.

Complication	Surgery
Distal / intramural stenosis	Antireflux reimplantation; balloon dilatation (restenosis common); meatoplasty for meatal stenosis
Long / multifocal stenosis	Boari flap; ileal ureter; ileal conduit; long-term nephrostomy
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

8 · Filariasis

Why this matters: in India, filariasis commonly causes hydrocele, funiculitis and chyluria.

- Organisms: *W. bancrofti* (>90%; *Culex* in urban India) and *Brugia*. **Only *W. bancrofti* causes genital filariasis**; *Brugia* affects the leg below the knee. Adults live 5–7 years; microfilariae are **nocturnally periodic**. Doxycycline kills the *Wolbachia* endosymbiont. Disease needs years of exposure and is rare before the teens.
- **Acute adenolymphangitis** is a **retrograde** lymphangitis (bacterial lymphangitis ascends instead). **Funiculoepididymitis** is a recurrent painful cord swelling sparing the testis, often mistaken for tumour, leading to needless orchidectomy.
- **Hydrocele** is the commonest chronic lesion (thick tunica, cholesterol or calcium, cord nodules). Elephantiasis comes late; lymph scrotum also occurs.

- Diagnosis: midnight Giemsa smear (positive in a minority); **circulating antigen** card (*W. bancrofti* only); IgG4 (exposure); US "**filarial dance sign**"; lymphoscintigraphy. "Burned-out" disease is test-negative.
- P3 Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria — *pyelolymphatic fistula; ether test, triglycerides, cystoscopic lateralisation, lymphoscintigraphy; MCT diet + DEC, then pelvic instillation sclerotherapy, then hilar lymphatic disconnection*

Not in the paper answer:

- Instil sclerosant on the side identified at cystoscopy.
- Most chyluria remits on conservative treatment.

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice: 6 mg/kg/day (2 mg/kg tds); a single-day course equals a 12-day course. **Contraindicated if onchocerciasis or *Loa loa* is possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg bd for 21 days; single dose in MDA
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with *Loa*
Doxycycline	Anti-*Wolbachia*; kills adults	200 mg daily for 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Treat all microfilaraemic patients. MDA: albendazole and ivermectin (Africa, where *Loa* or onchocerciasis is co-endemic) or albendazole and DEC (elsewhere, including India); triple therapy (IDA) is more suppressive.
- **Hydrocele** recurs after aspiration, so **excise the sac** and ligate leaking lymphatics. Funiculoepididymitis: excise nodules, preserve testis. Elephantiasis: reduction scrotoplasty; lymphadenectomy worsens drainage.
- Lymphoedema: hygiene, prevent bacterial infection, elevation, compression, physiotherapy.
- **Chyluria**: dilated retroperitoneal lymphatics rupture into the collecting system, giving milky urine with fat and protein loss; it is intermittent, may remit and precedes elephantiasis. Exclude TB, tumour and trauma.
- Onchocerciasis ("hanging groin"): ivermectin, then doxycycline; DEC blinds. Loiasis causes haematuria in ~30%.

9 · Hydatid disease and other parasites

Why this matters: blind puncture of a hydatid cyst causes anaphylaxis and seeding.

- P1 Hydatid disease of kidney — *kidney third site (2–3%); hydatiduria; water-lily sign, WHO stages; albendazole 400 mg bd 1–6 months; no aspiration; scolicide only if non-communicating; nephron-sparing surgery; PAIR not for kidney*

Not in the paper answer:

- Pack the field with hypertonic (20%) saline swabs; anaesthetist ready for anaphylaxis.
- Remove the cyst intact (pericystectomy) or do a partial nephrectomy; nephrectomy is only for a destroyed kidney.
- PAIR is avoided in the kidney (risk to the collecting system).
- The cyst grows slowly (1–2 cm/year), and renal function is usually preserved. Serology is adjunctive (60–90%). Perioperative albendazole (with or without PZQ) reduces seeding.

- **Enterobiasis** migrates ectopically in women (vulval granulomas, salpingitis), rarely to the epididymis. Give albendazole 400 mg or mebendazole 100 mg once, repeat at 2 weeks, and treat the household.
- **Amoebiasis**: renal involvement, painful genital ulcers. Give tinidazole or metronidazole, then paromomycin or iodoquinol.
- Trichomoniasis *(see Ch 29)*; fungal UTI *(see Ch 26)*.

★ Putting it together: infection decisions

A · UTI: drug matches site

Why this matters: urinary tract infection (UTI) drugs must reach the infected tissue.

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g once, TMP–SMX 3 days (resistance <20%), pivmecillinam	Narrow, urine-confined, flora-sparing
Pyelonephritis, complicated UTI, men	Tissue-penetrating (quinolone, cephalosporin; carbapenem if resistant)	Nitrofurantoin/fosfomycin do not reach tissue
ESBL producers	Fosfomycin/nitrofurantoin for cystitis; carbapenem if invasive	Co-resistance to quinolones and aminoglycosides
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching surgery	Otherwise harmful

B · Beyond antibiotics

Why this matters: obstruction, abscess and gas need drainage, not just antibiotics.

ALGORITHM

1. **Image** (ultrasound or CT): no response by 72 h, sepsis, obstruction/stone, gas, febrile men, recurrent pyelonephritis.
2. **Drain** any obstructed infected kidney (stent or nephrostomy); treat the stone later.
3. Renal abscess: if ≤3 cm, then antibiotics alone; 3–5 cm, variable; if >5 cm, then percutaneous drainage.
4. **Emphysematous pyelonephritis** (CT class): gas within kidney: antibiotics and drainage; if beyond it (poor prognosis), then nephrectomy (*see Ch 26*).
5. **Fournier:** antibiotics and **urgent repeated radical debridement**; scores never delay surgery.

C · Recurrent UTI in women

Why this matters: prophylaxis is stepped; antibiotics come last.

ALGORITHM

1. Confirm each episode by culture.
2. Behaviour, hydration; vaginal oestrogen if postmenopausal.
3. Methenamine, cranberry, D-mannose, immunostimulants.
4. Self-start or post-coital antibiotic; finally continuous low-dose prophylaxis.

D · Pelvic pain syndromes compared

Why this matters: both are diagnoses of exclusion.

	CP/CPPS	IC/BPS
Core	Male pelvic pain >3 months, no infection (NIH category III)	Filling-related bladder pain with LUTS, no other cause
Phenotyping	UPOINT; NIH-CPSI	Hunner vs non-Hunner (cystoscopy)

	CP/CPPS	IC/BPS
Treatment	By domain: α-blocker, antibiotic only if infection, pelvic-floor physiotherapy, neuromodulators, psychology	Behavioural, then oral (amitriptyline, PPS), then intravesical, then Hunner fulguration/injection, then BTX-A, neuromodulation; diversion last

E · GU tuberculosis in four lines

Why this matters: GU TB is easy to miss and scars as it heals.

ALGORITHM

1. **Suspect:** sterile pyuria, contracted bladder, beaded vas, epididymal mass.
2. **Confirm:** three early-morning urines: AFB, culture, GeneXpert.
3. **Treat:** 2 months HRZE, then 4 months HR; watch for ureteric stricture.
4. **Operate** after ≥4–6 weeks of drugs: nephrectomy (non-functioning), reimplantation (stricture), augmentation (contracted bladder) (*see Ch 31*).